

Machine Learning System

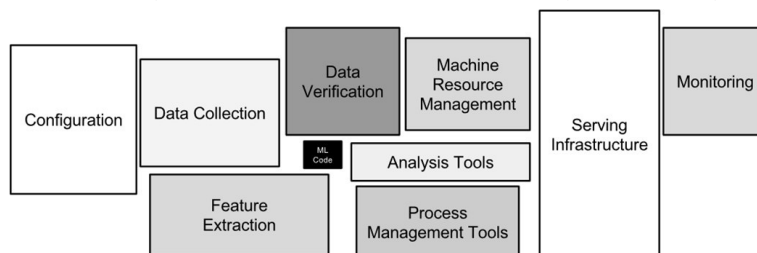
ONE LOVE. ONE FUTURE.

1

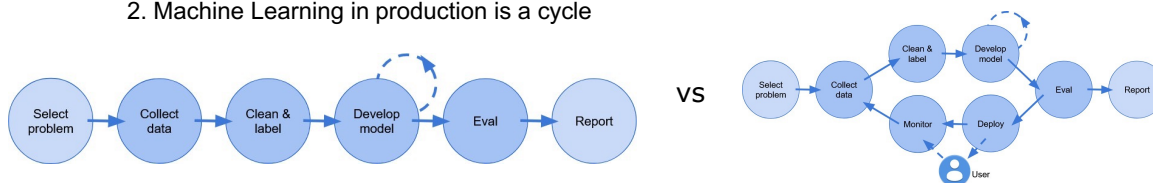
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Machine learning system

1. Machine Learning in production is much more than doing ML modelling



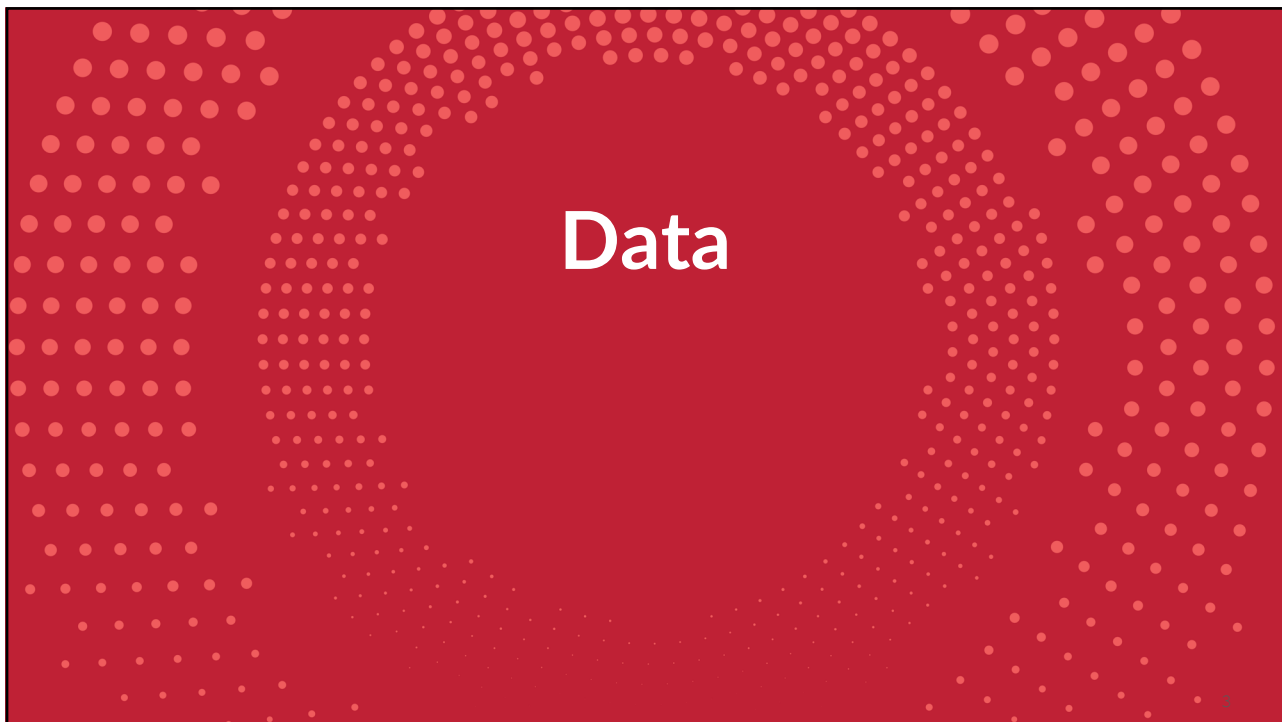
2. Machine Learning in production is a cycle



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Data is necessary.
The debate is whether *finite** data is sufficient.

* If we had infinite data, we can solve arbitrarily complex problems by just looking up the answers.

Massive data $\neq$ infinite data



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THE DATA SCIENCE HIERARCHY OF NEEDS

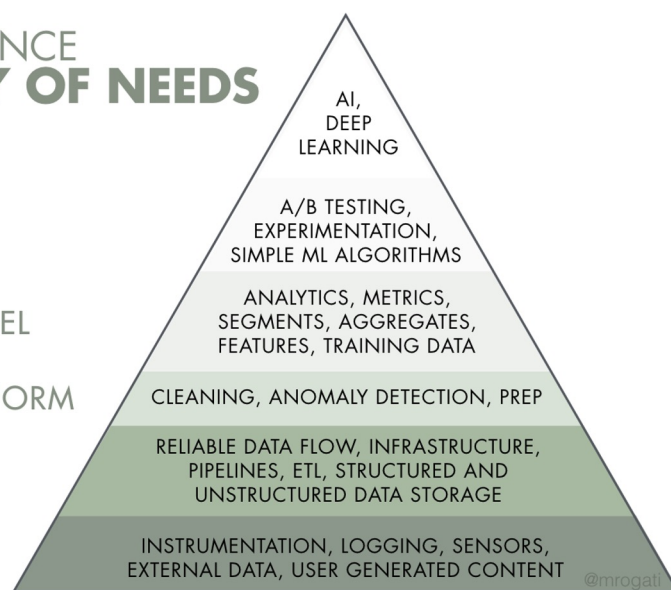
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

COLLECT



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[The AI Hierarchy of Needs](#) (Monica Rogati, 2017)

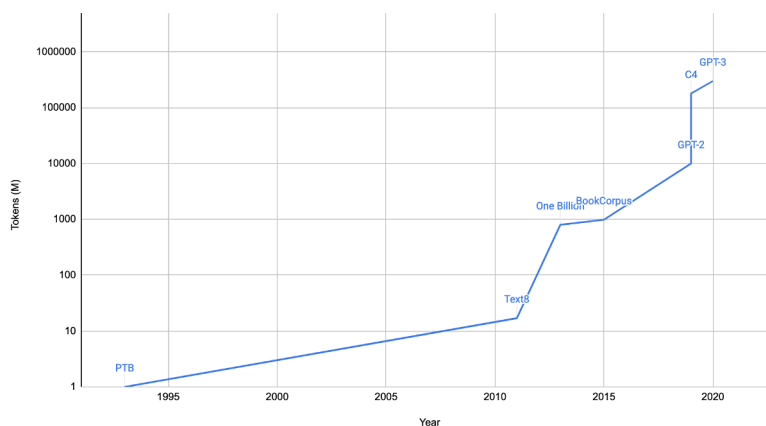
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Datasets for language models

Year	Tokens (M)	Dataset
1993	1	PTB
2011	17	Text8
2013	800	One Billion
2015	985	BookCorpus
2019	10,000	GPT-2
2019	180,000	C4
2020	300,000	GPT-3

Language model datasets over time (log scale)



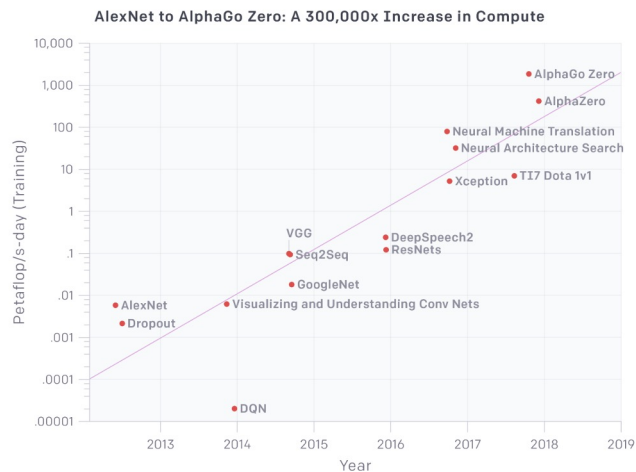
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More data (generally) needs more compute

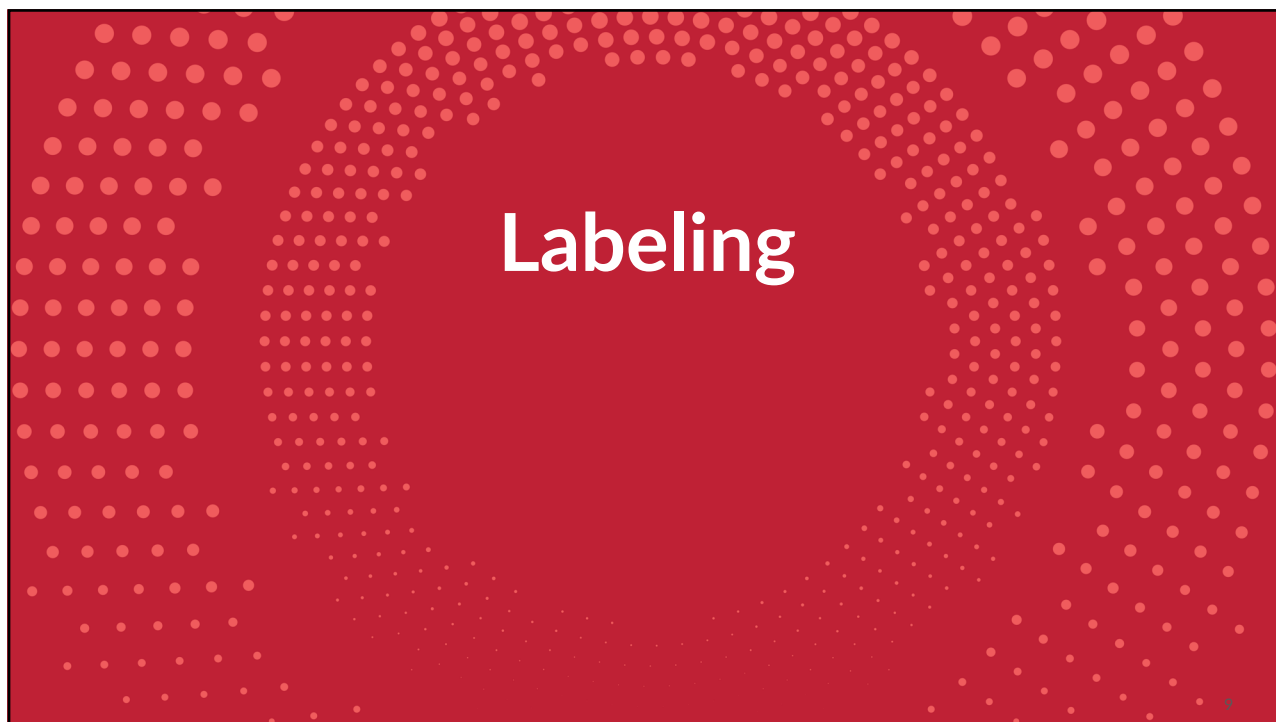
“amount of compute used in the largest AI training runs has doubled every 3.5 months”



⚠ Data: full of potential for biases ⚠

- sampling/selection biases
- under/over-representation of subgroups
- human biases embedded in historical data
- labeling biases
- ...

Algorithmic biases not covered (yet)!



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Labeling

1. Hand-labeling
2. Programmatic labeling
3. Weak supervision, semi supervision, active learning, transfer learning

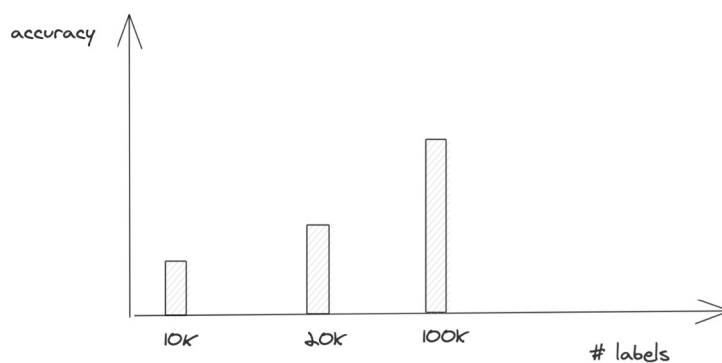


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! More data isn't always better !



🧠 Idea 🧠: crowdsource data to get 1 million labels!

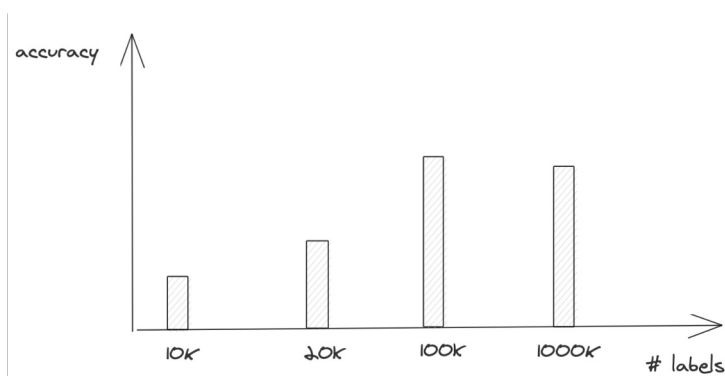


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! More data isn't always better !



Why is the model getting worse?

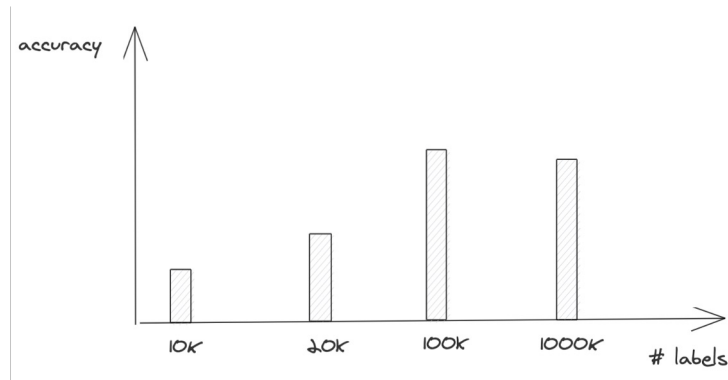


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⚠ Label sources with varying accuracy ⚠



- 100K labels: internally labeled, high accuracy
- 1M labels: crowdsourced, noisy



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Label multiplicity: example

Task: label all entities in the following sentence:

Darth Sidious, known simply as the Emperor, was a Dark Lord of the Sith who reigned over the galaxy as Galactic Emperor of the First Galactic Empire.



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Label multiplicity: example

Task: label all entities in the following sentence:

Darth Sidious, known simply as the Emperor, was a Dark Lord of the Sith who reigned over the galaxy as Galactic Emperor of the First Galactic Empire.

Annotator	# entities	Annotation
1	3	[Darth Sidious], known simply as the Emperor, was a [Dark Lord of the Sith] who reigned over the galaxy as [Galactic Emperor of the First Galactic Empire]
2	6	[Darth Sidious], known simply as the [Emperor], was a [Dark Lord] of the [Sith] who reigned over the galaxy as [Galactic Emperor] of the [First Galactic Empire].
3	4	[Darth Sidious], known simply as the [Emperor], was a [Dark Lord of the Sith] who reigned over the galaxy as [Galactic Emperor of the First Galactic Empire].



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which annotator is correct?

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Label multiplicity

More expertise required (more difficult to label), more room for disagreement!

If experts can't agree on a label, time to rethink human-level performance



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Label multiplicity: solution

- Clear problem definition
 - Pick the entity that comprises the longest substring

Annotator	# entities	Annotation
1	3	[Darth Sidious], known simply as the Emperor, was a [Dark Lord of the Sith] who reigned over the galaxy as [Galactic Emperor of the First Galactic Empire]
2	6	[Darth Sidious], known simply as the [Emperor], was a [Dark Lord] of the [Sith] who reigned over the galaxy as [Galactic Emperor] of the [First Galactic Empire].
3	4	[Darth Sidious], known simply as the [Emperor], was a [Dark Lord of the Sith] who reigned over the galaxy as [Galactic Emperor of the First Galactic Empire].



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Label multiplicity: solution

- Clear problem definition
- Annotation training
- Data lineage: track where data/labels come from



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Label multiplicity: solution

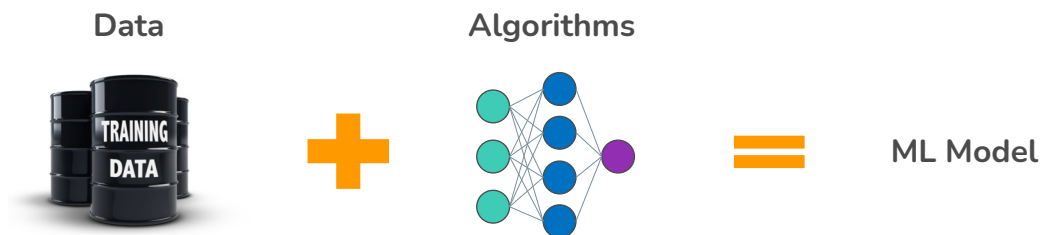
- Clear problem definition
- Annotation training
- Data lineage: track where data/labels come from
- Learning methods with noisy labels
 - [Learning with Noisy Labels](#) (Natarajan et al., 2013)
 - [Loss factorization, weakly supervised learning and label noise robustness](#) (Patrini et al., 2016)
 - [Cost-Sensitive Learning with Noisy Labels](#) (Natarajan et al., 2018)
 - [Confident Learning: Estimating Uncertainty in Dataset Labels](#) (Northcutt et al., 2019)



Programmatic labeling



Training data is the bottleneck

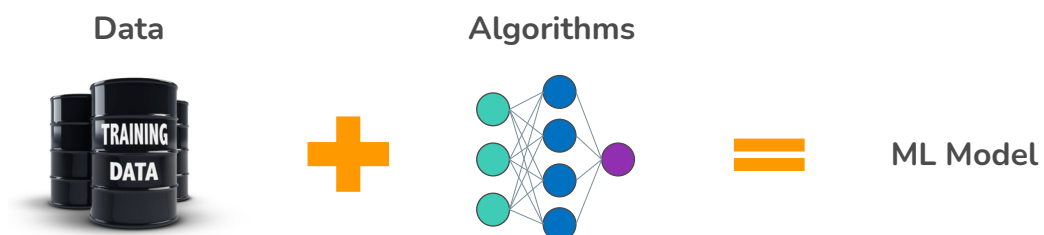


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Training data is the bottleneck



↑
Key differentiator

```
from transformers \
import BertModel as model
```

Increasingly
commoditized

"We don't have better algorithms. We just have more data."

Peter Norvig, [The Unreasonable Effectiveness of Data](#)

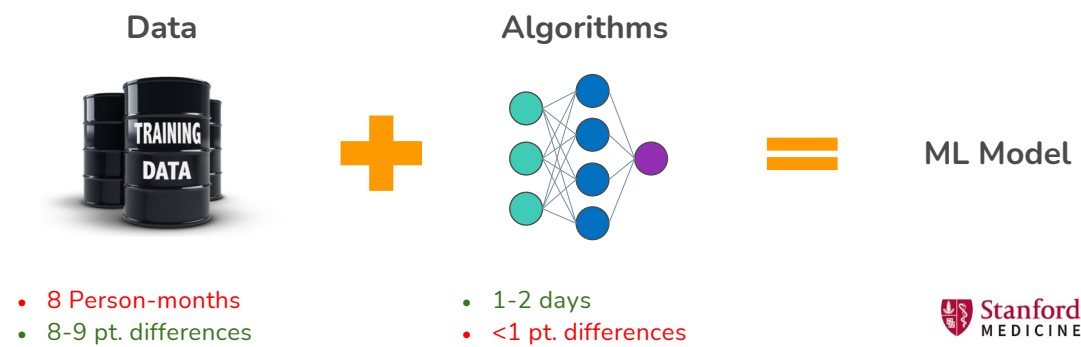


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Training data is the bottleneck



How to get training data in days?



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Cross-Modal Data Programming Enables Rapid Medical Machine Learning (Dunnmon et al., 2019)

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Hand labeling data is ...



- Expensive:** Esp. when subject matter **expertise** required
- Non-private:** Need to ship data to human annotators
- Slow:** Time required scales linearly with # labels needed
- Non-adaptive:** Every change requires re-labeling the dataset

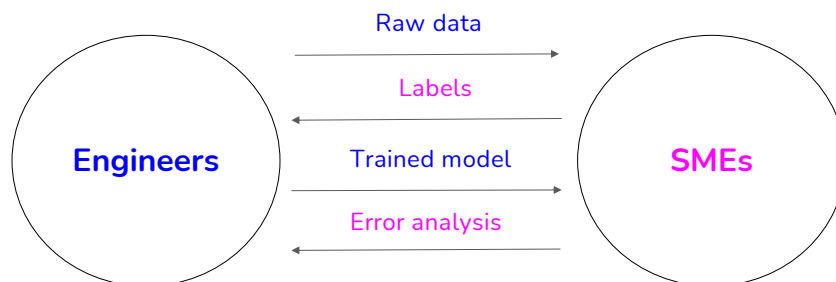


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Cross-functional communication



```
def function:
  if X:
    do Y
```

If the nurse's note mentions serious conditions like pneumonia, the patient's case should be given priority consideration.

Code: version control, reuse, share

How to version, share, reuse expertise?

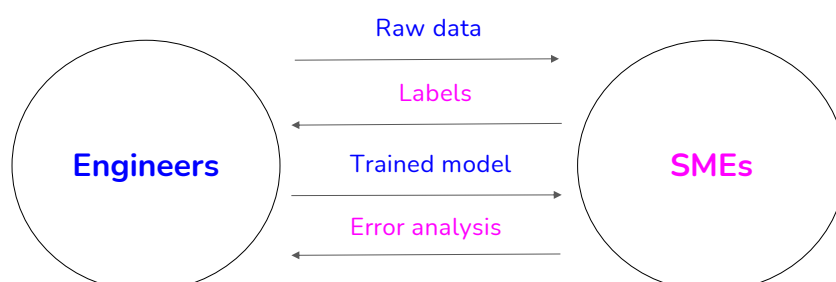


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SME as labeling functions



```
def function:
  if X:
    do Y
```

```
def labeling_function(note):
  if "pneumonia" in note:
    return "EMERGENT"
```

Labeling functions (LFs): Encode SME heuristics as functions and use them to label training data *programmatically*



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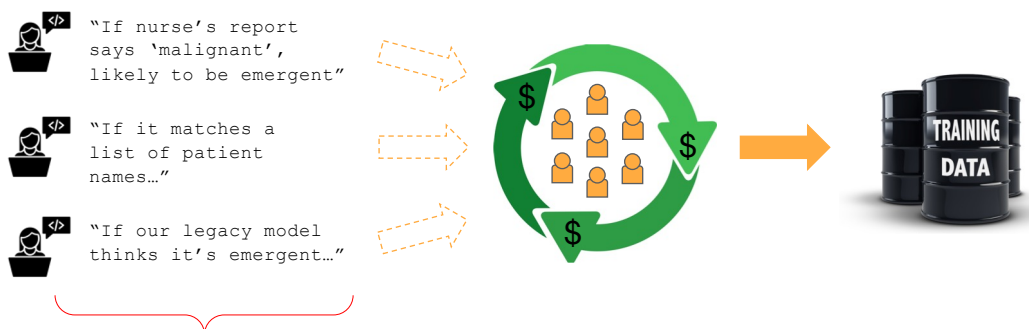
LFs: can express many different types of heuristics

Pattern Matching	If a phrase like "send money" is in email
Boolean Search	If <code>unknown_sender</code> AND <code>foreign_source</code>
DB Lookup	If sender is in our <code>Blacklist.db</code>
Heuristics	If <code>SpellChecker</code> finds 3+ spelling errors
Legacy System	If <code>LegacySystem</code> votes spam
Third Party Model	If <code>BERT</code> labels an entity "diet"
Crowd Labels	If <code>Worker #23</code> votes spam



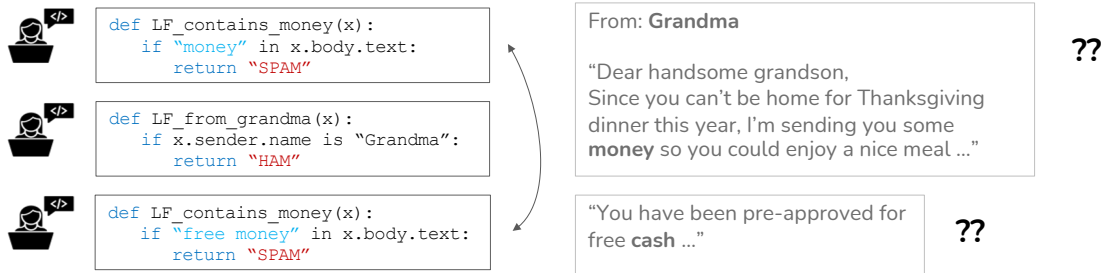
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LFs: can express many different types of heuristics



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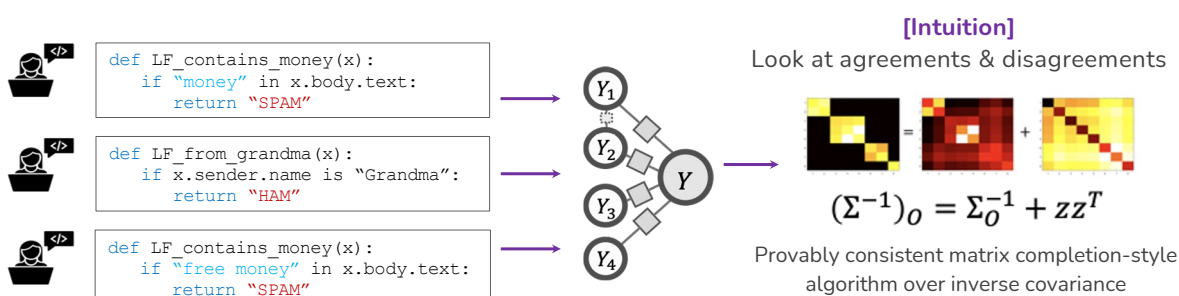
LFs: powerful but noisy



- **Noisy:** Unknown, inaccurate
- **Overlapping:** LFs may be correlated
- **Conflicting:** different LFs give different labels
- **Narrow:** Don't generalize well

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LF labels are combined to generate ground truths



[Ratner et. al. NeurIPS'16;
Bach et. al. ICML'17;
Ratner et. al. AAAI'19;
Varma et. al. ICML'19;
Sala et. al. NeurIPS'19;
Fu et. al. ICML'20]

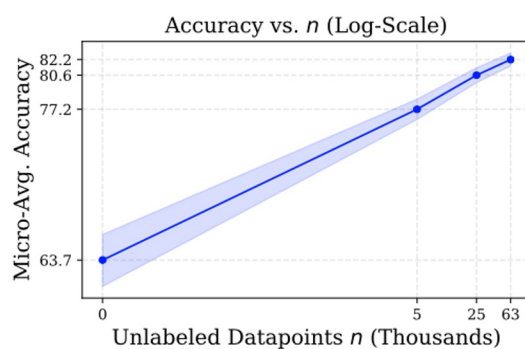
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Hand labeling	Programmatic labeling
Expensive: esp. when subject matter expertise required	Cost saving: Expertise can be versioned, shared, reused across organization
Non-private: Need to ship data to human annotators	Privacy: Create LFs using a cleared data subsample then apply LFs to other data without looking at individual samples.
Slow: Time required scales linearly with # labels needed	Fast: Easily scale 1K -> 1M samples
Non-adaptive: Every change requires re-labeling the dataset	Adaptive: When changes happen, just reapply LFs!



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Programmatic labeling: Scale with unlabeled data



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Weak supervision, semi-supervision, active learning, transfer learning

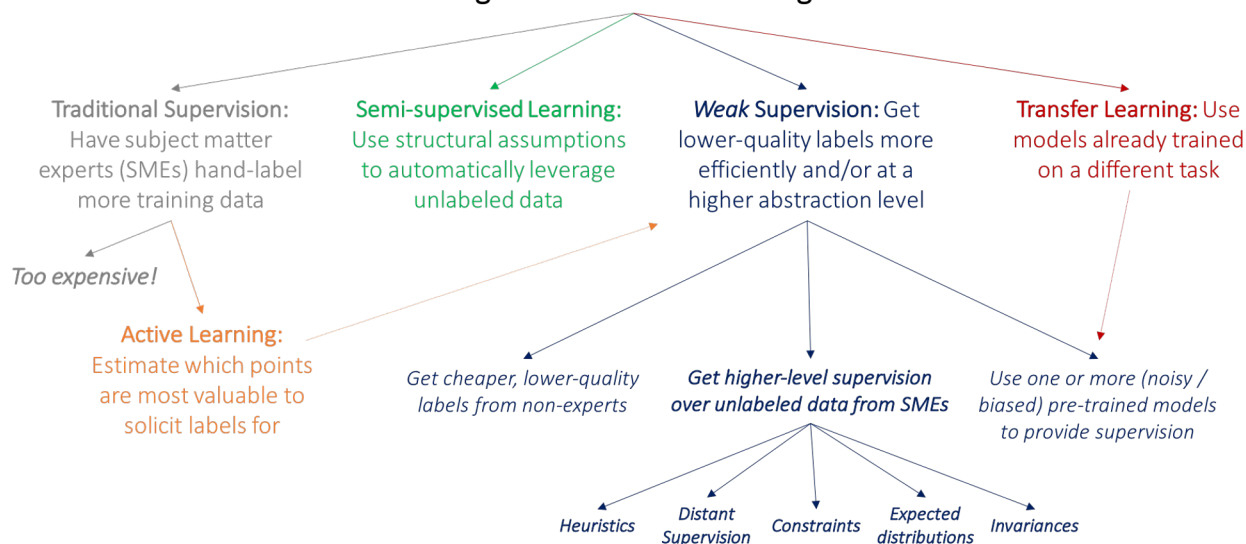


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How to get more labeled training data?



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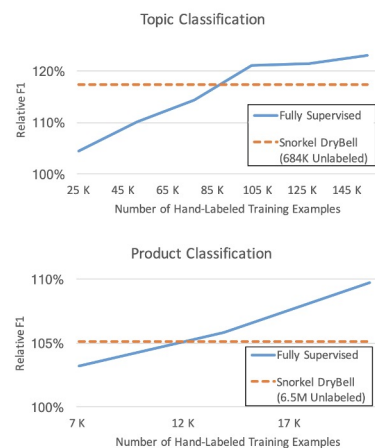
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Weak Supervision: A New Programming Paradigm for Machine Learning (Ratner et al., 2019)

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Weak supervision

- Leverage noisy, imprecise sources to create labels
 - e.g. if “money” is in an email it's probably spam



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Harnessing Organizational Knowledge for Machine Learning (Ratner et al., Google AI Blog 2019)

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Semi-supervision

- Use structural assumptions to leverage a large amount of unlabeled data together with a small amount of labeled data
 - Hashtags in the same profile/tweet are probably of similar topics



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Semi-supervision

- Use structural assumptions to leverage a large amount of unlabeled data together with a small amount of labeled data
- Might require complex algorithms like clustering to discover similarity



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Semi-supervision: self-training

1. Train model on a small set of labeled data
2. Use this model to generate predictions for unlabeled data
3. Use predictions with high raw probabilities as labels
4. Repeat step 1 with new labeled data



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Semi-supervision: perturbation-based methods

Assumption: small perturbation wouldn't change a sample's label

- Add white noises to images
- Add small values to word embeddings

Also a data augmentation method!



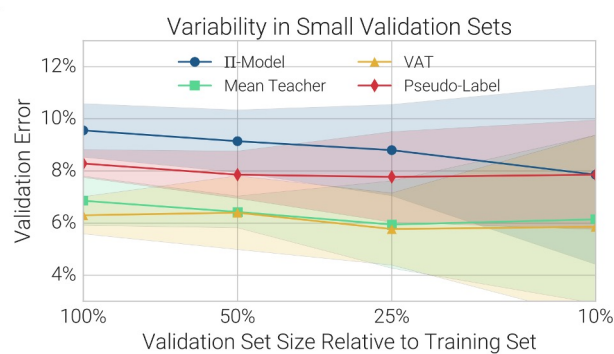
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Semi-supervision challenge: valid set's size

- Big valid set: less data for training
- Small valid set: not enough signal to choose the best model



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Realistic Evaluation of Deep Semi-Supervised Learning Algorithms (Oliver et al., 2018)

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Transfer learning

Apply model trained for one task to another task

1. Fine-tuning
2. Prompt-based



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[Language Models are Few-Shot Learners](#) (OpenAI 2020)

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Transfer learning: fine-tuning

- fine-tuning only some layers
- fine-tuning the entire model

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



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[Language Models are Few-Shot Learners](#) (OpenAI 2020)

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Transfer learning: Prompt-based

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese => ..... ← prompt
```



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Active learning

- **Goal:** Increase the efficiency of labels
- Label samples that are estimated to be most valuable to the model according to some metrics



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[Active Learning Literature Survey](#) (Burr Settles, 2010)

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Active learning metrics

- Uncertainty measurement
 - e.g. label samples with lowest raw probability for the predicted class



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[Active Learning Literature Survey](#) (Burr Settles, 2010)

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Active learning metrics

- Uncertainty measurement
- Candidate models' disagreement
 - Have several candidate models (e.g. models with different hyperparams)
 - Each model makes its own prediction
 - Label samples with most disagreement



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[Active Learning Literature Survey](#) (Burr Settles, 2010)

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Active learning

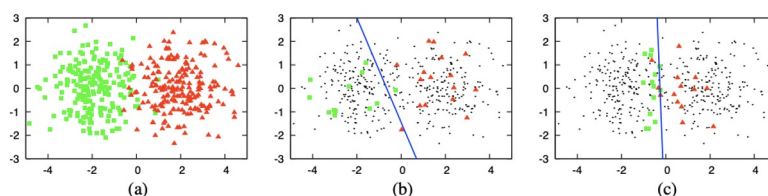


Figure 2: An illustrative example of pool-based active learning. (a) A toy data set of 400 instances, evenly sampled from two class Gaussians. The instances are represented as points in a 2D feature space. (b) A logistic regression model trained with 30 labeled instances randomly drawn from the problem domain. The line represents the decision boundary of the classifier (70% accuracy). (c) A logistic regression model trained with 30 actively queried instances using uncertainty sampling (90%).



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Active Learning Literature Survey (Burr Settles, 2010)

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Method	How	Ground truths required?
Weak supervision	Leverages (often noisy) heuristics to generate labels	No, but a small number of labels is useful to guide the development of heuristics
Semi-supervision	Leverages structural assumptions to generate labels	Yes. A small number of initial labels as seeds to generate more labels
Transfer learning	Leverages models pretrained on another task for your new task	No for zero-shot learning Yes for fine-tuning, though # GTs required is often much less than # GTs required if training from scratch.
Active learning	Labels data samples that are most useful to your model	Yes



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Sampling

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Types of sampling

- Non-probability sampling
 - Convenience sampling: selection based on availability
 - Soliciting response
 - Choosing existing datasets
 - Looking at available reviews on Amazon
 - Snowball sampling: future samples are selected based on existing samples
 - E.g. to scrape legit Twitter accounts, start with seed accounts then scrape their following
 - Judgment sampling: experts decide what to include
 - Quota sampling: quotas for certain slices of data (no randomization)
 -



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Data used in ML is mostly driven by convenience

- Language models: BookCorpus, CommonCrawl, Wikipedia, Reddit links
- Sentiment analysis: IMDB, Amazon
 - Only users who have access to the Internet and are willing to put reviews online
- Self-driving cars: most data is from the Bay Area (CA) and Phoenix (AZ)
 - Very little data on raining & snowing weather

⚠ Lots of biases in data! ⚠



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Types of sampling

- Non-probability sampling
- Random sampling
 - Simple random sampling
 - Stratified sampling
 - Weighted sampling
 - Importance sampling
 - Reservoir sampling
 - ...



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Simple random sampling

- Each sample in **population** has an equal chance of being selected
 - E.g. select 10% of all samples in population

Pros	Cons
Simple (easiest type of random sampling)	No representation guarantee: might exclude rare classes (black swan!)



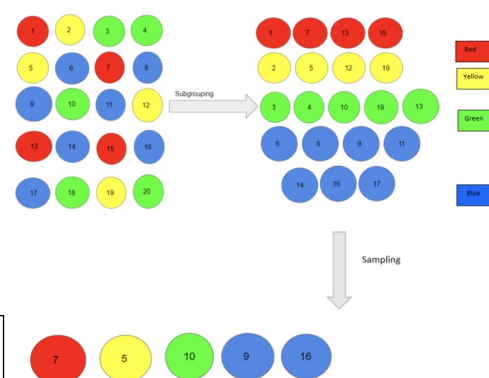
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Stratified sampling

- Divide population by subgroups
 - Slices of data
 - 20% of each age group: 18-24, 25-34, 35+, etc.
 - Classes
 - 2% of each class



Pros	Cons
Minor groups are represented	Can't be used when: - samples can't be put into subgroups - samples can belong in multiple subgroups (multilabel)



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Image from <https://www.analyticsvidhya.com/blog/2019/09/data-scientists-guide-8-types-of-sampling-techniques/>

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Weighted sampling

- Each element is given a weight, which determines the probability of being selected.
 - If you want to select a sample 30% of the time, give it 3/10 weight
- Might embed domain knowledge
 - E.g. know distribution of your target population or want to prioritize recent samples

```
random.choices(population=[1, 2, 3, 4, 100, 1000],
               weights=[0.2, 0.2, 0.2, 0.2, 0.1, 0.1],
               k=2)
```



```
random.choices(population=[1, 1, 2, 2, 3, 3, 4, 4, 100, 1000],
               k=2)
```



Importance sampling

- Useful when sampling from $P(x)$ is expensive, slow, or infeasible
 - Sample $x \sim Q(x)$ ← easier to sample from
 - Weight by $P(x)/Q(x)$
- ⚠ Calculating $P(x)/Q(x)$ might be expensive ⚠

$$E_{P(x)}[x] = \sum_x P(x) x = \sum_x Q(x) x \frac{P(x)}{Q(x)} = E_{Q(x)}\left[x \frac{P(x)}{Q(x)}\right]$$



Importance sampling

- Useful when sampling from $P(x)$ is expensive, slow, or infeasible
 - Sample $x \sim Q(x)$ ← easier to sample from
 - Weight by $P(x)/Q(x)$
- ⚠ Calculating $P(x)/Q(x)$ might be expensive ⚠
- E.g. essential for RL
 - Too expensive to estimate reward under new policy, so use old policy

$$E_{P(x)}[x] = \sum_x P(x) x = \sum_x Q(x) x \frac{P(x)}{Q(x)} = E_{Q(x)}\left[x \frac{P(x)}{Q(x)}\right]$$

$$\nabla_{\theta} J(\theta) = E_{\tau \sim \bar{\pi}_{\theta}(\tau)} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) \left(\prod_{t'=1}^t \frac{\pi_{\theta}(\mathbf{a}_{t'} | \mathbf{s}_{t'})}{\bar{\pi}_{\theta}(\mathbf{a}_{t'} | \mathbf{s}_{t'})} \right) \left(\sum_{t'=t}^T r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \right) \right]$$



use old policy to sample data

old policy

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Reservoir sampling: problem

- Need select k samples from a stream of n samples with equal probability
 - n is unknown
 - impossible/inefficient to fit all in memory
- Can stop the stream any moment and get the required samples

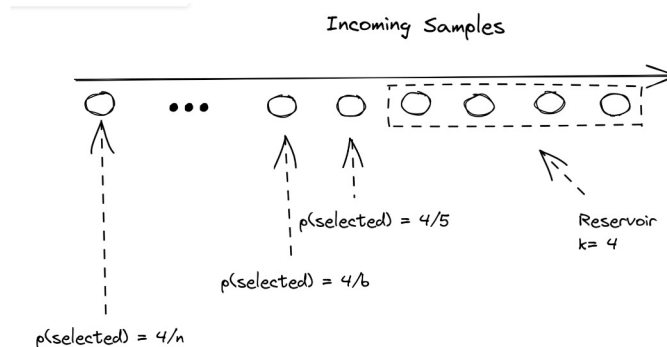


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Reservoir sampling: solution

1. First k elements are put in reservoir
2. For each incoming i^{th} element, generate a random number j between 1 and i
 - a. If $1 \leq j \leq k$: replace j^{th} in reservoir with i^{th}
3. Each incoming element has k/i chance of being in reservoir!



Also checkout algorithm L (based on geometric distribution)



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With vs. without replacement

With replacement	Without replacement
Same item can be chosen more than once	Same item can't be chosen more than once
• No covariance between two chosen samples • Approximate true population distribution	• Covariance between two chosen samples • Covariance reduced as dataset size becomes large
Bagging	Mini-batch gradient descent

Why do we use epochs instead of just sampling with replacement from the entire dataset?



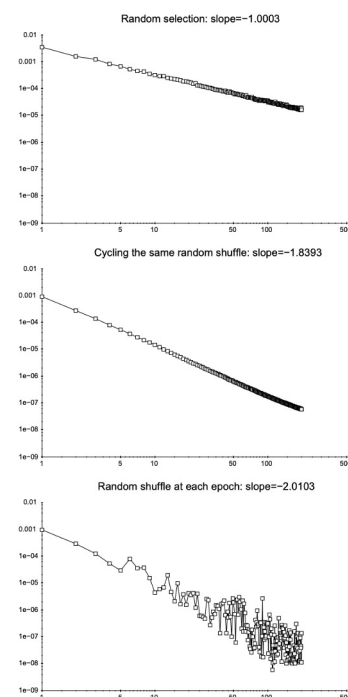
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With vs. without replacements

Because empirically it's converged faster
proven for strongly convex loss functions



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Curiously Fast Convergence of some Stochastic Gradient Descent Algorithms (Leon Bottou, 2009)

Why Random Reshuffling Beats Stochastic Gradient Descent (Gurbuzbalaba et al., 2015)

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Lab

<https://colab.research.google.com/github/GokuMohandas/Made-With-ML/blob/main/notebooks/madewithml.ipynb>



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Class imbalance

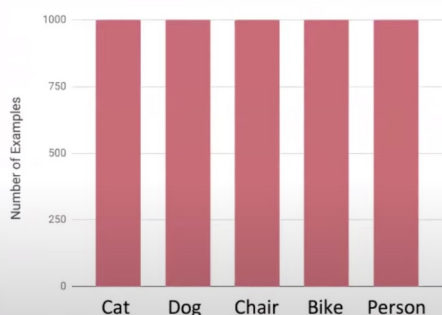
63

63

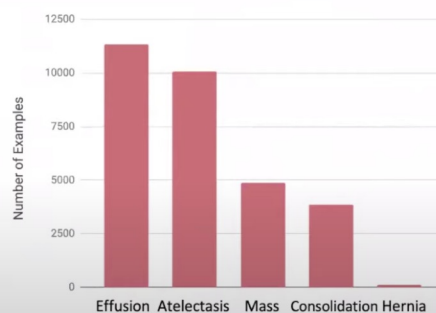
Class imbalance

Small data and rare occurrences

ML works well when
the data distribution
is this:



Not so well when it
is this:



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Andrew Ng: [Bridging AI's Proof-of-Concept to Production Gap](#) (2020)

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Why is class imbalance hard?

- Not enough signal to learn about rare classes



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Why is class imbalance hard?

- Not enough signal to learn about rare classes
- Statistically, predicting majority label has higher chance of being right
 - If a majority class accounts 99% of data, always predicting it gives 99% accuracy



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Why is class imbalance hard?

- Not enough signal to learn about rare classes
- Statistically, predicting majority label has higher chance of being right
- Asymmetric cost of errors: different cost of wrong predictions



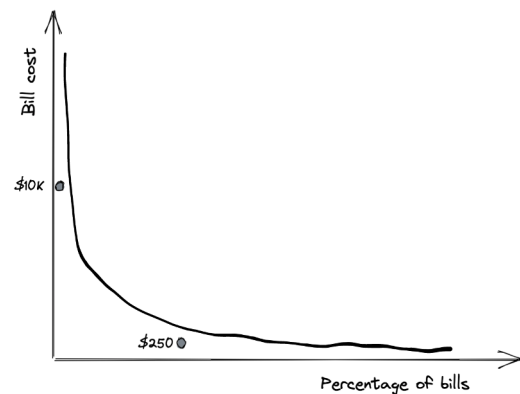
Why is class imbalance hard?

- Not enough signal to learn about rare classes
- Statistically, predicting majority label has higher chance of being right
- Asymmetric cost of errors: different cost of wrong predictions



Asymmetric cost of errors: regression

- 95th percentile: \$10K
- Median: \$250



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Thanks [Eugene Yan](#) for this example!

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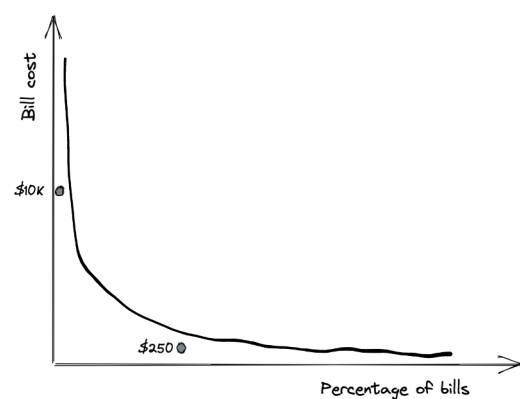
Asymmetric cost of errors: regression

100% error difference

- \$10K bill: off by \$10K
- \$250 bill: off by \$250

Not OK

OK



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Thanks [Eugene Yan](#) for this example!

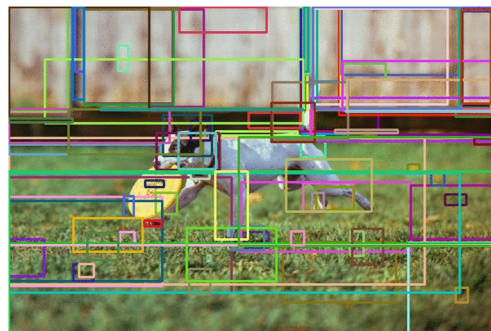
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Class imbalance is the norm

- Fraud detection
- Spam detection
- Disease screening
- Churn prediction
- Resume screening
 - E.g. 2% of resumes pass screening
- Object detection
 - Most bounding boxes don't contain any object

People are more interested in unusual/potentially catastrophic events



Sources of class imbalance

- Sampling biases
 - Narrow geographical areas (self-driving cars)
 - Selection biases
- Domain specific
 - Costly, slow, or infeasible to collect data of certain classes
- Labeling errors

How to deal with class imbalance

1. Choose the right metrics
2. Data-level methods
3. Algorithm-level methods



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1. Choose the right metrics

Model A vs. Model B confusion matrices

Zoom poll:
Which model would you choose?

Model A	Actual CANCER	Actual NORMAL
Predicted CANCER	10	10
Predicted NORMAL	90	890

Model B	Actual CANCER	Actual NORMAL
Predicted CANCER	90	90
Predicted NORMAL	10	810



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Choose the right metrics

Model A vs. Model B confusion matrices

Model B has a better chance of telling if you have cancer

Model A	Actual CANCER	Actual NORMAL
Predicted CANCER	10	10
Predicted NORMAL	90	890

Model B	Actual CANCER	Actual NORMAL
Predicted CANCER	90	90
Predicted NORMAL	10	810

Both have the same accuracy: 90%



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Symmetric metrics vs. asymmetric metrics

Symmetric metrics	Asymmetric metrics
Treat all classes the same	Measures a model's performance w.r.t to

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$$

- TP: True positives
- TN: True negatives

- FP: False positives
- FN: False negatives



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Class imbalance: asymmetric metrics

- Your model's performance w.r.t to a class

	CANCER (1)	NORMAL (0)	Accuracy	Precision	Recall	F1
Model A	10/100	890/900	0.9	0.5	0.1	0.17
Model B	90/100	810/900	0.9	0.5	0.9	0.64

⚠ F1 score for CANCER as 1
is different from F1 score for
NORMAL as 1 ⚠



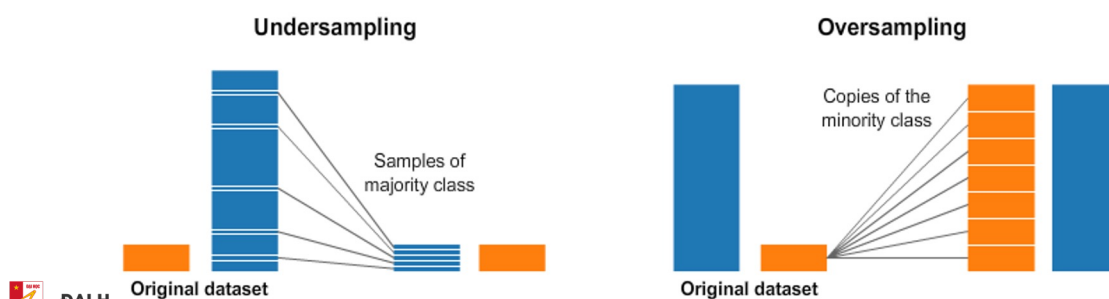
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2. Data-level methods: Resampling

Undersampling	Oversampling
Remove samples from the majority class	Add more examples to the minority class



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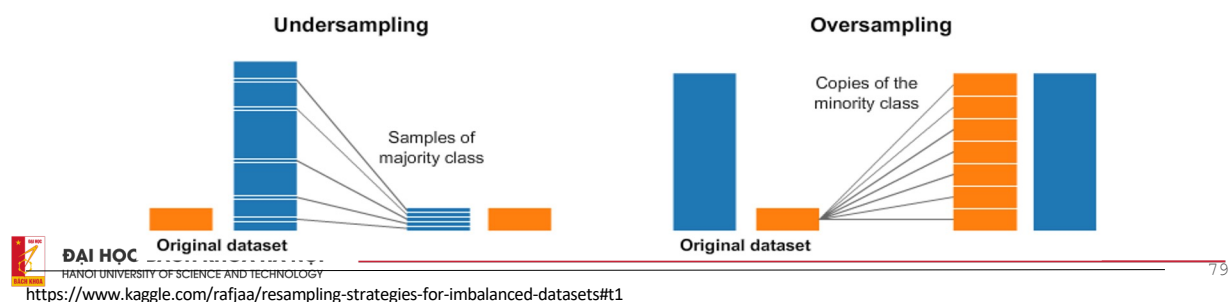
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<https://www.kaggle.com/rafjaa/resampling-strategies-for-imbalanced-datasets#t1>

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2. Data-level methods: Resampling

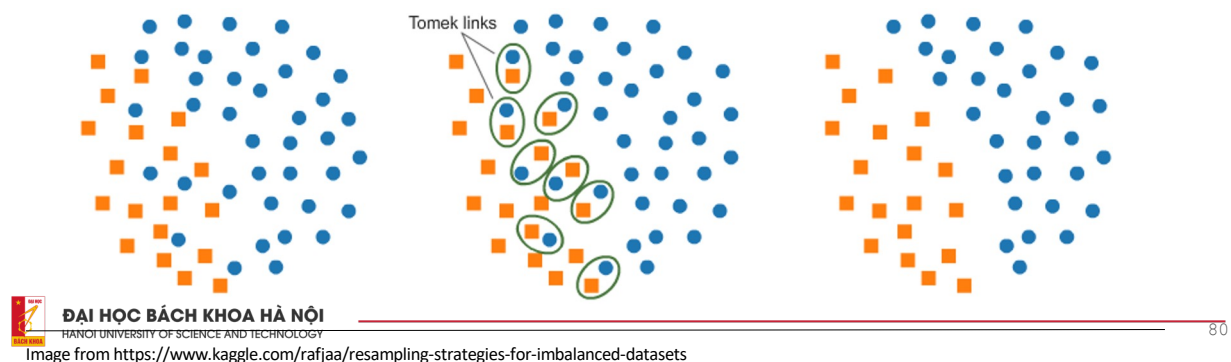
Undersampling	Oversampling
Remove samples from the majority class	Add more examples to the minority class
Can cause overfitting	Can cause loss of information



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Undersampling: Tomek Links

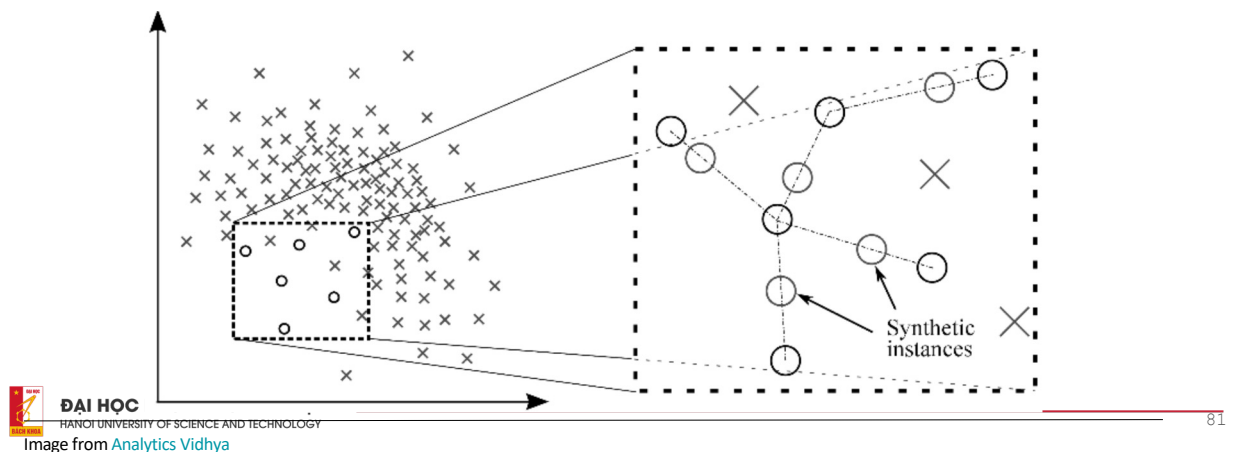
- Find pairs of close samples of opposite classes
- Remove the sample of majority class in each pair
 - Pros: Make decision boundary more clear
 - Cons: Make model less robust



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Oversampling: SMOTE

- Synthesize samples of minority class as convex ($\sim$ linear) combinations of existing points and their nearest neighbors of same class.

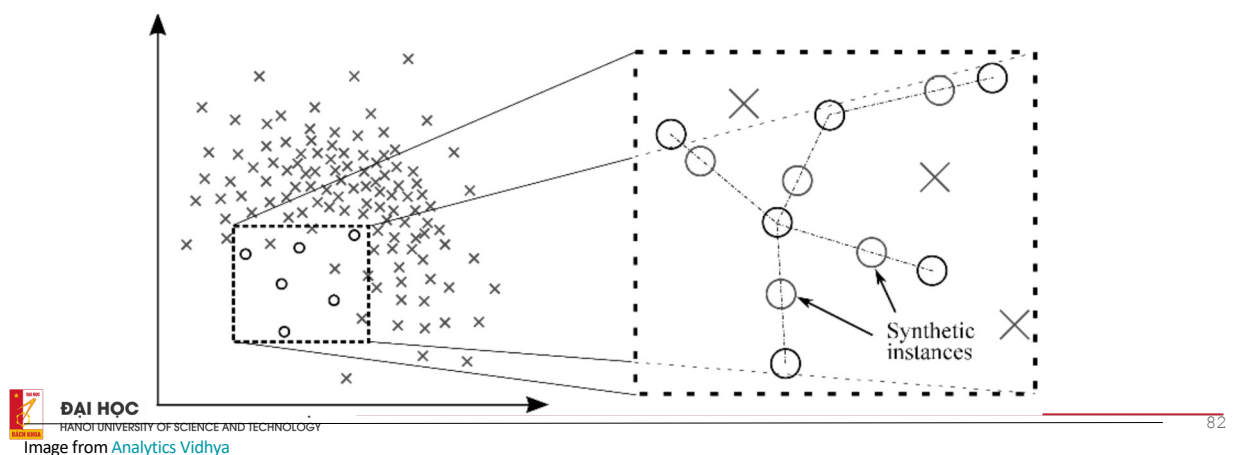


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Oversampling: SMOTE

Both SMOTE and Tomek links only work on low-dimensional data!

- Synthesize samples of minority class as convex ($\sim$ linear) combinations of existing points and their nearest neighbors of same class.



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3. Algorithm-level methods

- Naive loss: all samples contribute equally to the loss
- Idea: training samples we care about should contribute more to the loss

$$L(X; \theta) = \sum_x L(x; \theta)$$



3. Algorithm-level methods

- Cost-sensitive learning
- Class-balanced loss
- Focal loss



Cost-sensitive learning

- C_{ij} : the cost if class i is classified as class j

	Actual NEGATIVE	Actual POSITIVE
Predicted NEGATIVE	$C(0, 0) = C_{00}$	$C(1, 0) = C_{10}$
Predicted POSITIVE	$C(0, 1) = C_{01}$	$C(1, 1) = C_{11}$

- The loss caused by instance x of class i will become the weighted average of all possible classifications of instance x .

$$L(x; \theta) = \sum_j C_{ij} P(j | x; \theta)$$



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Class-balance loss

- Give more weight to rare classes

Non-weighted loss

$$L(X; \theta) = \sum_i L(x_i; \theta)$$

Weighted loss

$$L(X; \theta) = \sum_i W_{y_i} L(x_i; \theta)$$

$$W_c = \frac{N}{\text{number of samples of class } C}$$

```
model.fit(features, labels, epochs=10, batch_size=32, class_weight={"fraud": 0.9, "normal": 0.1})
```



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Focal loss

- Give more weight to difficult samples:
 - downweights well-classified samples

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{otherwise,} \end{cases}$$

$$\text{CE}(p_t) = -\log(p_t)$$

$$\text{FL}(p_t) = -(1 - p_t)^\gamma \log(p_t)$$



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Focal Loss for Dense Object Detection (Lin et al., 2017)

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Lab

<https://colab.research.google.com/github/GokuMohandas/Made-With-ML/blob/main/notebooks/madewithml.ipynb>



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Feature engineering

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Data augmentation

“Data augmentation is the
new feature engineering”

- Josh Wills, prev Director of Data Engineering @ Slack



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Data augmentation: goals

- Improve model's performance overall or on certain classes
- Generalize better
- Enforce certain behaviors



Data augmentation

1. Simple label-preserving transformation
2. Perturbation
3. Data synthesis



Label-preserving: Computer Vision

Random cropping, flipping, erasing, etc.

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Image-to-Image Pre-training Attention Network
for Person Re-Identification (Huang et al., 2021)



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Label-preserving: NLP

Original sentences	I'm so happy to see you.
Generated sentences	I'm so glad to see you. I'm so happy to see y'all . I'm very happy to see you.

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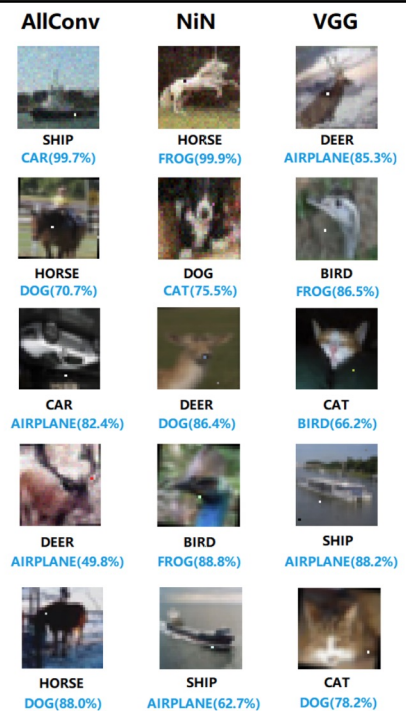
Perturbation: neural networks can be sensitive to noise

- 67.97% Kaggle CIFAR-10 test images
- 16.04% ImageNet test images

can be misclassified by changing just one pixel
([Su et al., 2017](#))



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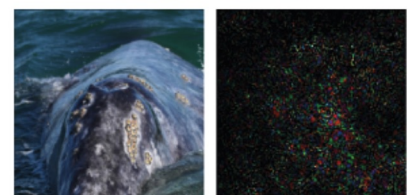
Perturbation: Computer Vision

- Random noise
- Search strategy
 - [DeepFool](#) (Moosavi-Dezfooli et al., 2016): find the minimal noise injection needed to cause a misclassification with high confidence.

Whale



Turtle
noise by
DeepFool



Turtle
noise by fast
gradient sign



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Perturbation: NLP

- Random replacement
 - e.g. BERT ($10\% * 15\% = 1.5\%$)
- 80% of the time: Replace the word with the [MASK] token, e.g., my dog is hairy → my dog is [MASK]
- 10% of the time: Replace the word with a random word, e.g., my dog is hairy → my dog is apple
- 10% of the time: Keep the word unchanged, e.g., my dog is hairy → my dog is hairy. The purpose of this is to bias the representation towards the actual observed word.



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BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (Devlin et al., 2018)

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Data synthesis: NLP

- Template-based
 - Very common in conversational AI
- Language model-based

Template	Find me a [CUISINE] restaurant within [NUMBER] miles of [LOCATION].
Generated queries	• Find me a Vietnamese restaurant within 2 miles of my office. • Find me a Thai restaurant within 5 miles of my home. • Find me a Mexican restaurant within 3 miles of Google headquarters.



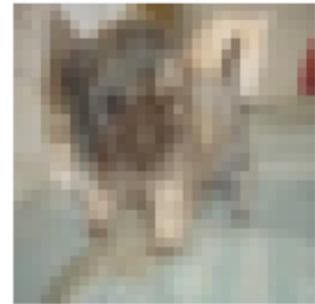
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Data Synthesis: Computer Vision

- Mixup
 - Create convex combination of samples of different classes
 - Labels: cat [3], dog [4]
 - Mixup: 30% dog, 70% cat [$0.3 * 3 + 0.7 * 4 = 3.7$]



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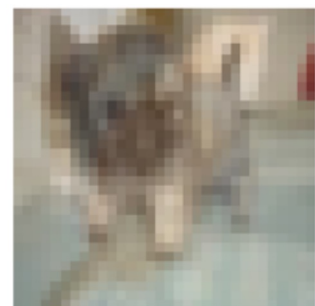
[mixup: Beyond Empirical Risk Minimization](#) (Zhang et al., 2017)

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Data Synthesis: Computer Vision

- Mixup
 - Incentivize models to learn linear relationships
 - Improves generalization on speech and tabular data
 - Can be used to stabilize the training of GANs



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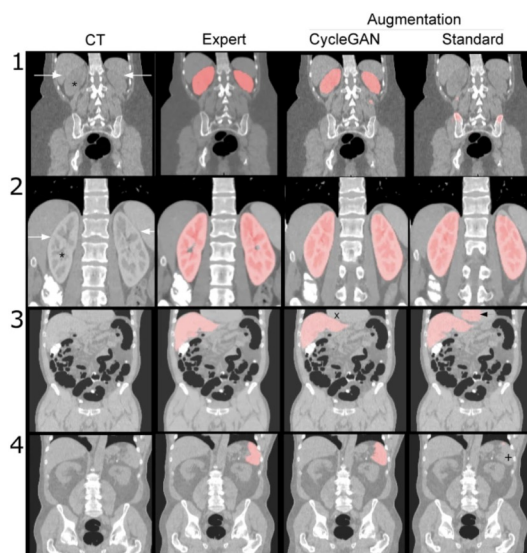
<https://forums.fast.ai/t/mixup-data-augmentation/22764>

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Data augmentation: GAN

Example: kidney segmentation with data augmentation by CycleGAN



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Data augmentation using generative adversarial networks (CycleGAN) to improve generalizability in CT segmentation tasks (Sandfort et al., 2019)

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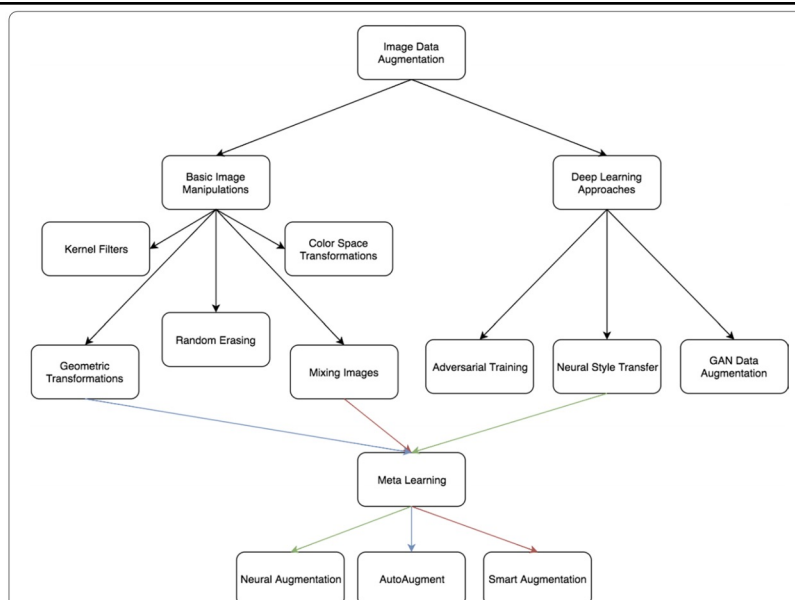


Fig. 2 A taxonomy of image data augmentations covered; the colored lines in the figure depict which data augmentation method the corresponding meta-learning scheme uses, for example, meta-learning using Neural Style Transfer is covered in neural augmentation [36]



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A survey on Image Data Augmentation for Deep Learning (Connor Shorten & Taghi M. Khoshgohar, 2019)

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Learned features vs. engineered features



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Feature engineering

Wait, doesn't deep learning promise no more feature engineering?



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Feature engineering

Wait, doesn't deep learning promise no more feature engineering?

- We're still very far from that point
- Many ML models in industry aren't deep learning



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Engineered features: text

Stopword removal

I have a dog. He's sleeping.

Lemmatization

I have dog. He's sleep~~ing~~.

Contraction

I have dog. He's sleep.

Punctuation

I have dog. He is sleep.

Lowercase

I have dog He is sleep

N-gram

i have dog he is sleep

features

0	3	0	0	...
1	0	0	0	...
0	0	1	0	..
...	...	...	...	...

samples



Classifier
(e.g. LogReg)



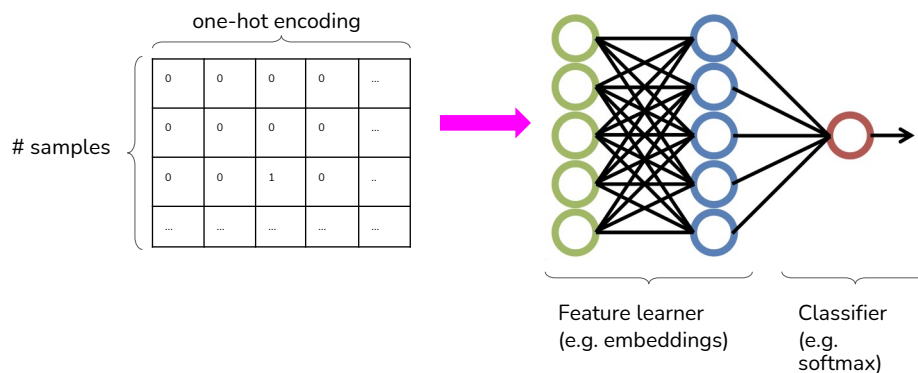
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I	you	have	dog	cat	he	she	is	they	sleep	I, have	have, dog	good, dog	...
1	0	1	0	1	0	1	0	1	1	1	1	0	...

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Learned features: text



Text I have a dog. He's sleeping.

One-hot	I	you	have	dog	cat	he	she	is	they	sleep	mom	food	yes	...
	1	0	1	1	0	1	0	1	0	0	0	0	0	...



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Learned features: spam classification

Comment ID	Time	User	Text	# ▲	# ▼	Link	# img	Thread ID	Reply to	# replies	...
93880839	2020-10-30 T 10:45 UTC	gitrekt	Your mom is a nice lady.	1	0	0	0	2332332	n0tab0t	1	...

User ID	Created	User	Subs	# ▲	# ▼	# replies	Karma	# threads	Verified email	Awards	...
4402903	2015-01-57 T 3:09 PST	gitrekt	[r/m/, r/memes, r/socialist]	15	90	28	304	776	No		...

Thread ID	Time	User	Text	# ▲	# ▼	Link	# img	# replies	# views	Awards	...
93883208	2020-10-30 T 2:45 PST	doge	Human is temporary, AGI is forever	120	50	1	0	32	2405	1	...



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Feature engineering: spam classification

Even more features:

- Post frequency, max posts per day
- Post repetitiveness
- Language detection, typos, abnormal punctuations, ratio uppercase/lowercase
- IP, other users from the same IP
- NSFW words, blacklisted links
- Targeted users
- ...



Feature engineering

- For complex tasks, number of features can go up to millions!
- Lots of ML production work involves coming up with new features
 - Fraudsters come up with new techniques very fast, so need to come up with new features very fast to counter
- Often require subject matter expertise



Common feature engineering ops

1. Handling missing values
2. Scaling
3. Discretization
4. Categorical features
5. Feature crossing
6. Positional embeddings



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Handling missing values

- Not all missing values are equal
 - Missing not at random (MNAR)
 - Missing at random (MAR)
 - Missing completely at random (MCAR)



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Handling missing values

Missing not at random – when a value is missing due to the value itself

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1		A	150,000		1	Engineer	No
2	27	B	50,000			Teacher	No
3		A	100,000	Married	2		Yes
4	40	B	(\$350,000?)		2	Engineer	Yes
5	35	B	(\$350,000?)	Single	0	Doctor	Yes
6		A	50,000		0	Teacher	No
7	33	B	60,000	Single		Teacher	No
8		B	10,000			Student	No

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Handling missing values

Missing at random – when a value is missing due to another observed variable

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1		A	150,000		1	Engineer	No
2	27	B	50,000			Teacher	No
3		A	100,000	Married	2		Yes
4	40	B			2	Engineer	Yes
5	35	B		Single	0	Doctor	Yes
6		A	50,000		0	Teacher	No
7	33	B	60,000	Single		Teacher	No
8		B	10,000			Student	No

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Handling missing values

Missing completely at random – there is no pattern to which values are missing

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1		A	150,000		1	Engineer	No
2	27	B	50,000			Teacher	No
3		A	100,000	Married	2		Yes
4	40	B			2	Engineer	Yes
5	35	B		Single	0	Doctor	Yes
6		A	50,000		0	Teacher	No
7	33	B	60,000	Single		Teacher	No
8	26	B	10,000			Student	No

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Handling missing values

- Deletion – removing data with missing entries
- Imputation – filling missing fields with certain values

Many people prefer deletion not because
it's better, but it's easier to do

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Handling missing values

- Deletion
 - Column deletion – remove columns with too many missing entries
 - drawbacks – even if half the values are missing, the remaining data still potentially useful information for predictions
 - e.g. even if over half the column for 'Marital status' is missing, marital status is still highly correlated with house purchasing
 - Row deletion

Marital status
Married
Single
Single



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Handling missing values

- Deletion
 - Column deletion
 - Row deletion



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Handling missing values

- Row deletion
 - Good for: data missing completely at random (MCAR) and few values missing

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1	39	A	150,000	Married	1	Engineer	No
2	27	B	50,000	Single	0	Teacher	No
3		A	100,000	Married	2		Yes
4	40	B	75,000	Married	2	Engineer	Yes
5	35	B	35,000	Single	0	Doctor	Yes
6	32	A	50,000	Married	0	Teacher	No
7	33	B	60,000	Single	2	Teacher	No
8	20	B	10,000	Single	1	Student	No



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Handling missing values

- Row deletion
 - Bad when many examples have missing fields

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1		A	150,000		1	Engineer	No
2	27	B	50,000			Teacher	No
3		A	100,000	Married	2		Yes
4	40	B			2	Engineer	Yes
5	35	B		Single	0	Doctor	Yes
6		A	50,000		0	Teacher	No
7	33	B	60,000	Single		Teacher	No
8	20	B	10,000			Student	No



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Handling missing values

- Row deletion
 - Bad for: missing values are not at random (MNAR)
 - Missing information is information itself

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1		A	150,000		1	Engineer	No
2	27	B	50,000			Teacher	No
3		A	100,000	Married	2		Yes
4	40	B	(\$350,000?)		2	Engineer	Yes
5	35	B	(\$350,000?)	Single	0	Doctor	Yes
6		A	50,000		0	Teacher	No
7	33	B	60,000	Single		Teacher	No
8	20	B	10,000			Student	No

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Handling missing values

- Row deletion
 - Bad for: missing data at random (MAR)
 - Can potentially bias data – we've accidentally removed all examples with gender 'A'

ID	Age	Gender	Annual income	Marital status	Number of children	Job	Buy?
1		A	150,000		1	Engineer	No
2	27	B	50,000			Teacher	No
3		A	100,000	Married	2		Yes
4	40	B			2	Engineer	Yes
5	35	B		Single	0	Doctor	Yes
6		A	50,000		0	Teacher	No
7	33	B	60,000	Single		Teacher	No
8	20	B	10,000			Student	No

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Imputation

- Fill missing fields with certain values
 - Defaults
 - E.g. 0, or the empty string, etc.
 - Statistical measures – mean, median, mode
 - e.g. if a day in July is missing its temperature value, fill it with the median temperature in July



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Imputation

- Fill missing fields with certain values
 - Defaults
 - E.g. 0, or the empty string, etc.
 - Statistical measures – mean, median, mode
 - e.g. if a day in July is missing its temperature value, fill it with the median temperature in July

Avoid filling missing values with possible values!



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Scaling



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Types of scaling

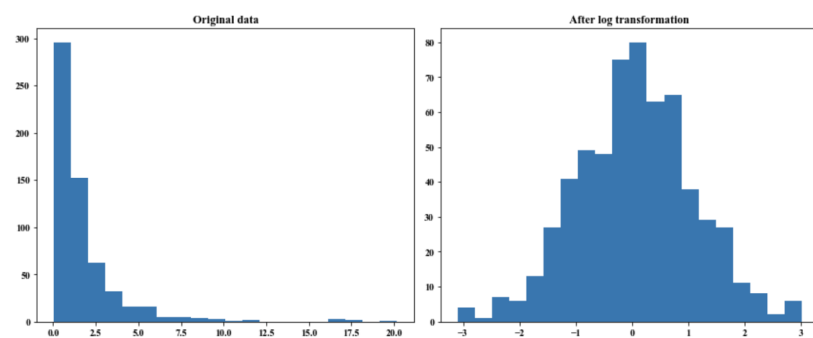
scaling type	use case
min/max normalization	Any -- no assumptions about variables
z-score normalization	When variables follow a normal distribution
log scaling	When variables follow an exponential distribution



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Log scaling

- Help with skewed data
- Often gives performance gain



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Scaling

scaling type	use case
min/max normalization	Any -- no assumptions about variables
z-score normalization	When variables follow a normal distribution
log scaling	When variables follow an exponential distribution

- scaling can be a common source of data leakage
- scaling variables requires global statistics



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Discretization

- Turning a continuous feature into a discrete feature (quantization)



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Discretization

- Turning a continuous feature into a discrete feature (quantization)
- Create buckets for different ranges
 - Incorporate knowledge/expertise about each variable by constructing specific buckets



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Discretization

- Turning a continuous feature into a discrete feature (quantization)
- Create buckets for different ranges
 - Incorporate knowledge/expertise about each variable by constructing specific buckets
- Examples
 - Income
 - Lower income: $x < \$35,000$
 - Middle income: $\$35,000 < x < \$100,000$
 - High income: $x > \$100,000$



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Discretization

- Turning a continuous feature into a discrete feature (quantization)
- Create buckets for different ranges
 - Incorporate knowledge/expertise about each variable by constructing specific buckets
- Examples
 - Income
 - Lower income: $x < \$35,000$
 - Middle income: $\$35,000 \leq x < \$100,000$
 - High income: $x \geq \$100,000$
 - Age
 - Minors: $x < 18$
 - College: $18 \leq x < 22$
 - Young adult: $22 \leq x < 30$
 - 30 ≤ x < 40
 - 40 ≤ x < 65
 - Seniors: $x \geq 65$



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Encoding Categorical Features

- Example: you want to build a recommendation system for Amazon
 - There are over 2 million brands that we need to recommend



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Encoding Categorical Features

How do we encode the different brands/vendors?



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Encoding Categorical Features

- one-hot encoding!

How do we encode the different brands/vendors?



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Encoding Categorical Features

- one-hot encoding!

How do we handle a new brand that wants to join Amazon?



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Encoding Categorical Features

- one-hot encoding!
- encode unseen brands with “UNKNOWN”

How do we handle a new brand that wants to join Amazon?



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Encoding Categorical Features

- one-hot encoding!
- encode unseen brands with “UNKNOWN”

Problem! “UNKNOWN” was not seen during training, so none of the products in this category are being recommended



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Encoding Categorical Features

- one-hot encoding!
- encode unseen brands with “UNKNOWN”

Fix – encode brands as themselves, group bottom-performing
1% of brands as “UNKNOWN”



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Encoding Categorical Features

- one-hot encoding!
- encode unseen brands with “UNKNOWN”
- Group bottom 1% of brands and newcomers into “UNKNOWN” category



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Encoding Categorical Features

- one-hot encoding!
- encode unseen brands with “UNKNOWN”
- Group bottom 1% of brands and newcomers into “UNKNOWN” category

Alert! Nike wants to join Amazon as a new vendor



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Encoding Categorical Features

- one-hot encoding!
- encode unseen brands with “UNKNOWN”
- Group bottom 1% of brands and newcomers into “UNKNOWN” category
- Problem – this treats all newcomers the same as unpopular brands on the platform



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Encoding Categorical Features

How do we implement a flexible method of handling new brands as they are introduced to our system?



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Encoding New Categories

1. Represent each category with its attribute
 - a. E.g. to represent a brand, use features: yearly revenue, company size, etc..
2. Hashing trick



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Encoding Categorical Features

- Hashing – use a hash function to hash categories to different indexes



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Encoding Categorical Features

- Hashing – use a hash function to hash categories to different indexes
 - e.g. $\text{hash}(\text{"Nike"}) = 0$, $\text{hash}(\text{"Adidas"}) = 27$, etc...



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Encoding Categorical Features

- Hashing – use a hash function to hash categories to different indexes
 - e.g. $\text{hash}(\text{"Nike"}) = 0$, $\text{hash}(\text{"Adidas"}) = 27$, etc...
- Benefits – you can choose how large the hash space is



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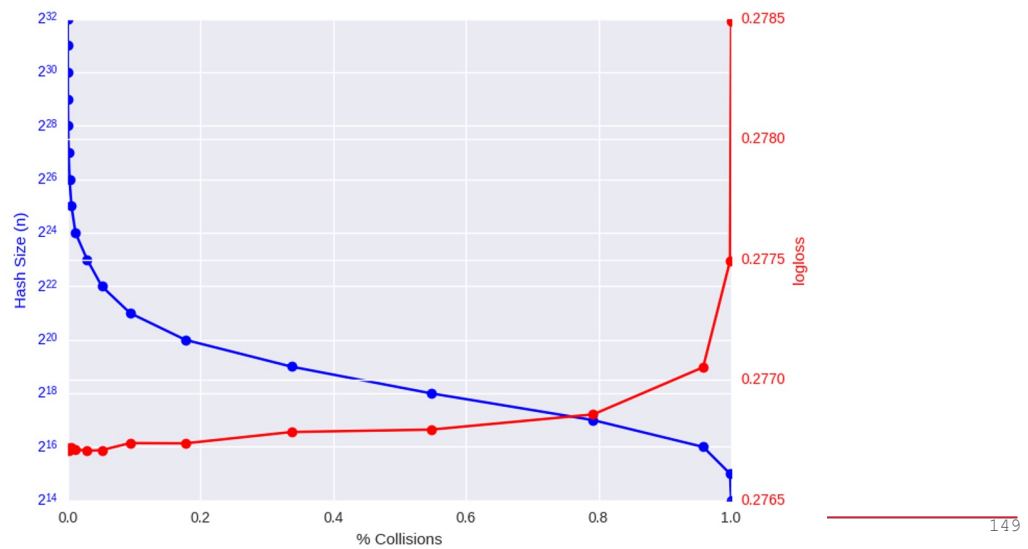
Encoding Categorical Features

- Hashing – use a hash function to hash categories to different indexes
 - e.g. $\text{hash}(\text{"Nike"}) = 0$, $\text{hash}(\text{"Adidas"}) = 27$, etc...
- Benefits – you can choose how large the hash space is
- Drawbacks – two categories being hashed to the same index



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Encoding Categorical Features



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Encoding Categorical Features

- Choose a hash space large enough to reduce collisions
- Choose functions with properties beneficial to your use case
 - Locality-sensitive hashing



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Hashing Trick Takeaways

- Hashing trick considered “hacky” by academics
- Widely used in industry and in machine learning frameworks
- Useful in practice for continual learning in production



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Feature Crossing

- Combine two or more features to create a new feature

Marriage	Single	Married	Single	Single	Married
Children	0	2	1	0	1
Marriage & children	Single, 0	Married, 2	Single, 1	Single, 0	Married, 1



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Feature Crossing

- Helps models learn non-linear relationships between variables
- **Warning** – feature crossing can blow up your feature space
 - e.g. Feature A and B both have 100 categories → Feature A x B will have 10,000 categories
 - Need even more data to learn this new feature space
 - Blowing up feature space can increase risk of overfitting

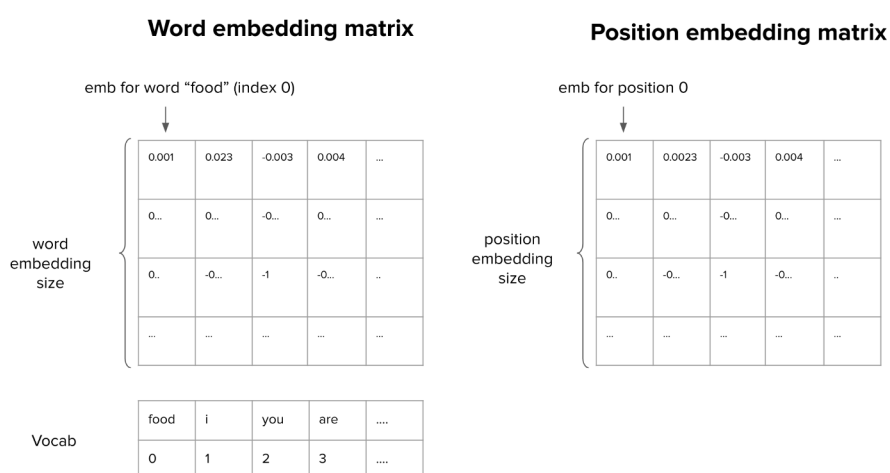
Very common in RecSys & CTR with models like [DeepFM](#) and [xDeepFM](#)



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Positional Embeddings

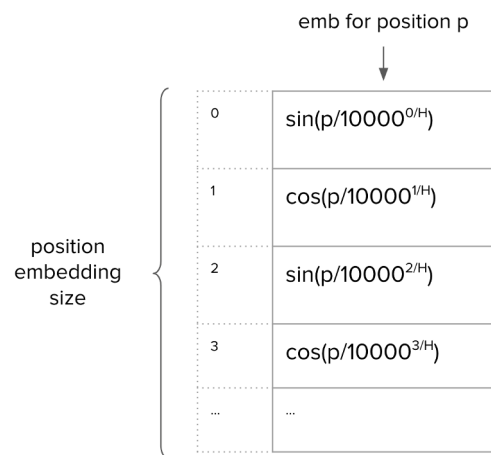
- Popularized in [Attention is All You Need](#) paper
- Similar to word embeddings
 - Can be either learned or fixed



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Positional Embeddings

- Fourier features



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Positional Embeddings

Why do we need position embeddings?



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Positional Embeddings

Why do we need position embeddings?

- Traditional architectures (RNNs, LSTMs) process tokens sequentially
- Transformers process tokens in parallel → need to communicate sequential nature of human language to model



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Lab

<https://colab.research.google.com/github/GokuMohandas/Made-With-ML/blob/main/notebooks/madewithml.ipynb>



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Data leakage



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Data leakage

- Some form of the label “leaks” into the features
- This same information is not available during inference



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Data leakage: example 1

- Problem: detect lung cancer from CT scans
- Data: collected from hospital A
- Performs well on test data from hospital A
- Performs poorly on test data from hospital B

Patient ID	Date	Doctor note	Medical record	Scanner type	CT scan
------------	------	-------------	----------------	--------------	---------



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Data leakage: example 1

- Problem: detect lung cancer from CT scans
- Data: collected from hospital A
- Performs well on test data from hospital A
- Performs poorly on test data from hospital B

Patient ID	Date	Doctor note	Medical record	Scanner type	CT scan
------------	------	-------------	----------------	--------------	---------

At hospital A, when doctors suspect that a patient has lung cancer, they send that patient to a higher-quality scanner



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Data leakage: example 2

- Problem: predicting how many views an article will get
- Data: historical data on the site
- Where might data leakage come from?

Article ID	Date	Title	Article	Author	Language	Translations
------------	------	-------	---------	--------	----------	--------------



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Data leakage: example 2

- Problem: predicting how many views an article will get
- Data: historical data on the site

Not leakage because author popularity also available during inference

Article ID	Date	Title	Article	Author	Language	Translations
------------	------	-------	---------	--------	----------	--------------

The site only translate articles that are already gaining attention



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Data leakage: Kaggle edition



What will be done about data leaks?

Louka Ewington-Pitsos · Last comment 2Y ago by Gunes Evitan

23

7 comments ...



Time for Kaggle to clarify

bluetrain · Last comment 2Y ago by Vishal

56

8 comments ...



There is something new about Leaks ?

Haythem Teillii · Last comment 2Y ago by Oskin Nikita

4

1 comment ...



New leakage web finded

Zhang Yunfei · Last comment 2Y ago by Zhang Yunfei

19

19 comments ...



Zidmie

Topic Author

3rd place

The leak explained!

Posted in [liverpool-lon-switching](#) 2 years ago

60

The cat5 data (category with 10 open channels) is very similar to the addition of two signals of cat4 data, as several teams noticed. But also, we can find that the data from 4000001 to 4100000 (cat4) was used to create the data from 5700001 to 5800000 : $\text{openchannels}(5700001:5800000) = \text{openchannels}(4000001 \text{ to } 4100000) + \text{openchannels of other cat4 data (that I didn't really look for)}$. So we can just subtract data $\text{openchannels}(4000001 \text{ to } 4100000)$ from the private LB data. Then we have cat4 data, much easier to predict. By using this method, we just reached a score of 0.9543. I guess by digging further, it can lead to the score of 0.985!

[ASHRAE - Great Energy Predictor III](#)

[University of Liverpool - lon Switching](#)

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Causes of data leakage

1. Splitting time-correlated data randomly instead of by time



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Partition: shuffle then split

	Week 1	Week 2	Week 3	Week 4	Week 5
Test split	X11	X21	X31	X41	X51
Valid split	X12	X22	X32	X42	X52
Train split	X13	X23	X33	X43	X53
	X14	X24	X34	X44	X54
	...	...	...	...	...

Aim for similar distributions of labels across splits
 e.g. each split has 90% NEGATIVE, 10% POSITIVE



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Partition: shuffle then split

	Week 1	Week 2	Week 3	Week 4	Week 5
Test split	X11	X21	X31	X41	X51
Valid split	X12	X22	X32	X42	X52
Train split	X13	X23	X33	X43	X53
	X14	X24	X34	X44	X54
	...	...	...	...	...

⚠ Not representative of real-world usage! ⚠



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Partition: shuffle then split

	Week 1	Week 2	Week 3	Week 4	Week 5
Test split	X11	X21	X31	X41	X51
Valid split	X12	X22	X32	X42	X52
Train split	X13	X23	X33	X43	X53
	X14	X24	X34	X44	X54
	...	...	...	...	...

A source of data leakage. Examples:

- stock price prediction
- song recommendation



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A better partition

	Train split				
	Week 1	Week 2	Week 3	Week 4	Week 5
	X11	X21	X31	X41	X51
	X12	X22	X32	X42	X52
	X13	X23	X33	X43	X53
	X14	X24	X34	X44	X54
	...	...	...	...	...

Valid split

Test split



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Solution: split data by time

Train split					
Week 1	Week 2	Week 3	Week 4	Week 5	
X11	X21	X31	X41	X51	Valid split
X12	X22	X32	X42	X52	
X13	X23	X33	X43	X53	Test split
X14	X24	X34	X44	X54	
...	...	...	...	...	

Also forces you to think about the cold-start problem



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Causes of data leakage

1. Splitting time-correlated data randomly instead of by time
2. Data processing before splitting
 - a. Use the whole dataset (including valid/test) to generate global statistics/info



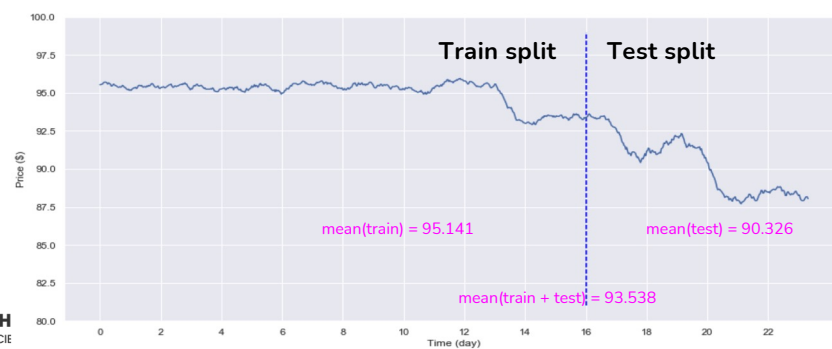
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2. Data processing before splitting

- Use the whole dataset (including valid/test) to generate global statistics/info
 - mean, variance, min, max, n-gram count, vocabulary, etc.
- Statistics are then used to process test data
 - scale, fill in missing values, etc.



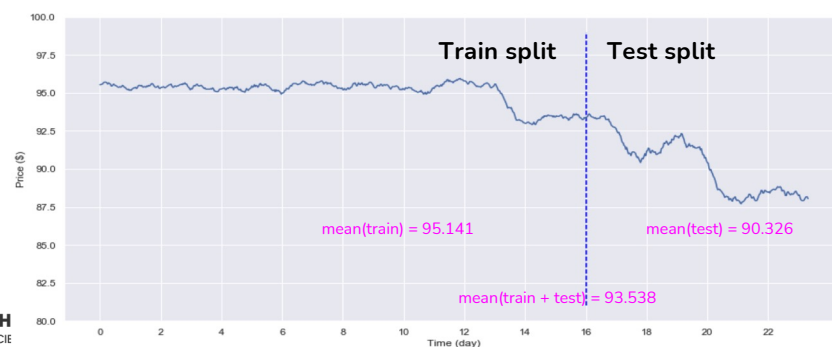
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2. Data processing before splitting

- Use the whole dataset (including valid/test) to generate global statistics/info
- Solution:
 - Split your data before scaling/filling in missing values
 - Split even before any EDA to ensure you're blind to the test set



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Causes of data leakage

1. Splitting time-correlated data randomly instead of by time
2. Data processing before splitting
3. Poor handling of data duplication before splitting
 - a. Test set includes data from the train set



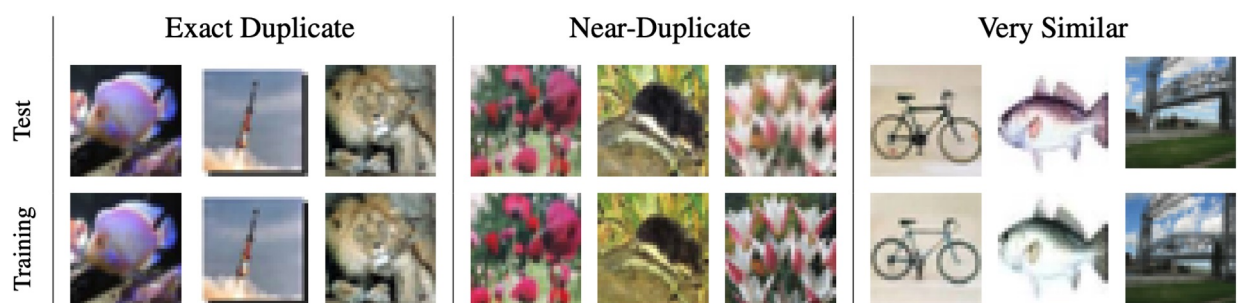
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3. Poor handling of data duplication before splitting

- Datasets come with duplicates & near-duplicates
 - 3.3% CIFAR-10 and 10% CIFAR-100 test images have dups in training set
 - Removing dups increases errors 17.05% -> 19.38% on CIFAR-100 [PyramidNet-272-200]



Do we train on test data? Purging CIFAR of near-duplicates (Barz & Denzler, 2019)

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3. Poor handling of data duplication before splitting

- Datasets come with duplicates & near-duplicates
- Oversampling can cause duplications



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3. Poor handling of data duplication before splitting

- Test set includes data from the train set
- Solution:
 - Deduplicate data before splitting
 - Oversample after splitting



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Causes of data leakage

1. Splitting time-correlated data randomly instead of by time
2. Data processing before splitting
3. Poor handling of data duplication before splitting
4. Group leakage
 - a. A group of examples have strongly correlated labels but are divided into different splits



Causes of data leakage

1. Splitting time-correlated data randomly instead of by time
2. Data processing before splitting
3. Poor handling of data duplication before splitting
4. Group leakage
 - a. A group of examples have strongly correlated labels but are divided into different splits
 - b. Example: CT scans of the same patient a week apart
 - c. **Solution: Understand your data and keep track of its metadata**



Causes of data leakage

1. Splitting time-correlated data randomly instead of by time
2. Data processing before splitting
3. Poor handling of data duplication before splitting
4. Group leakage
5. Leakage from data generation & collection process
 - a. Example: doctors send high-risk patients to a better scanner
 - b. Solution: Data normalization + subject matter expertise



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Causes of data leakage

1. Splitting time-correlated data randomly instead of by time
2. Data processing before splitting
3. Poor handling of data duplication before splitting
4. Group leakage
5. Leakage from data generation & collection process



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How to detect leakage?



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How to detect leakage?

1. Measure correlation of a feature with labels
 - a. A feature alone might not cause leakage, but 2 features together might



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How to detect leakage?

1. Measure correlation of a feature with labels
2. Feature ablation study
 - a. If removing a feature causes the model performance to decrease significantly, figure out why.



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How to detect leakage?

1. Measure correlation of a feature with labels
2. Feature ablation study
3. Monitor model performance as more features are added
 - a. Sudden increase: either a very good feature or leakage!



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How to engineer good features



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Evaluating a feature

1. Feature importance
2. Feature generalization



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Measuring a feature's importance

How much the model performance deteriorates
if a feature or a set of features containing that feature
is removed from the model?



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Measuring a feature's importance

- XGBoost

```
get_score(fmap="", importance_type='weight')
```

Get feature importance of each feature. For tree model Importance type can be defined as:

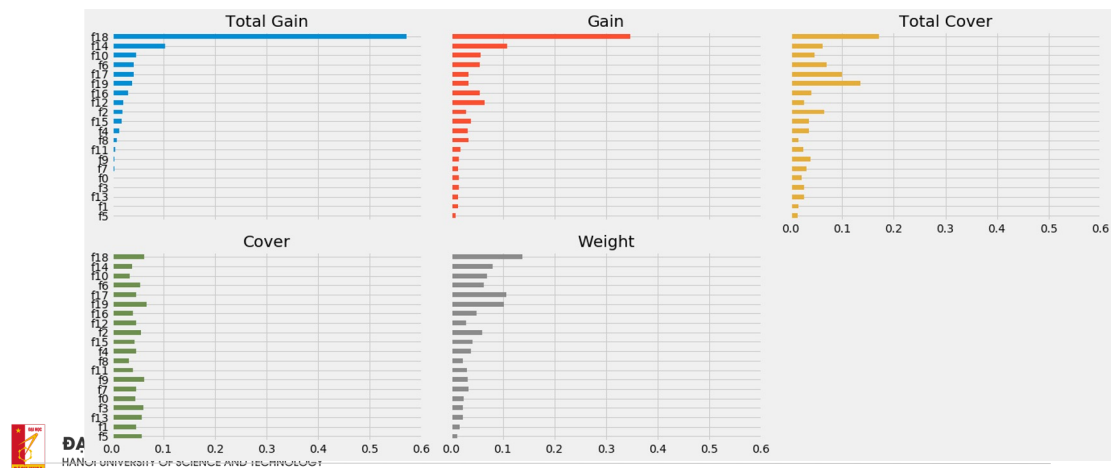
- 'weight': the number of times a feature is used to split the data across all trees.
- 'gain': the average gain across all splits the feature is used in.
- 'cover': the average coverage across all splits the feature is used in.
- 'total_gain': the total gain across all splits the feature is used in.
- 'total_cover': the total coverage across all splits the feature is used in.



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Measuring a feature's importance

- XGBoost



The Multiple faces of 'Feature importance' in XGBoost (Amjad Abu-Rmileh, 2019)

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SHAP: SHapley Additive exPlanations

- Measuring a feature's contribution to a single prediction

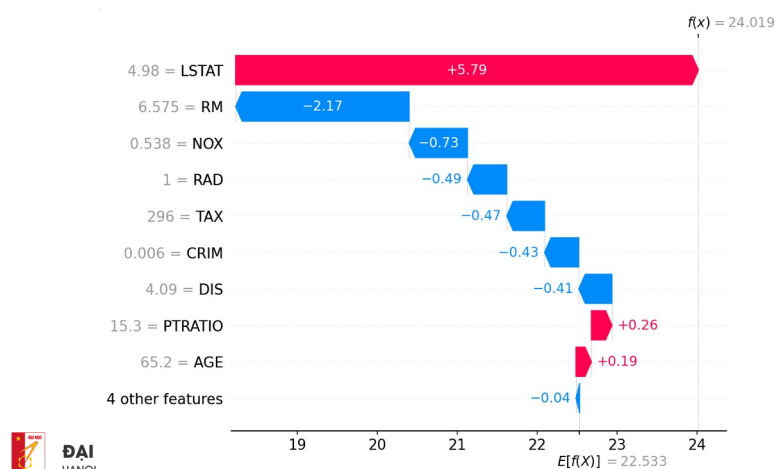


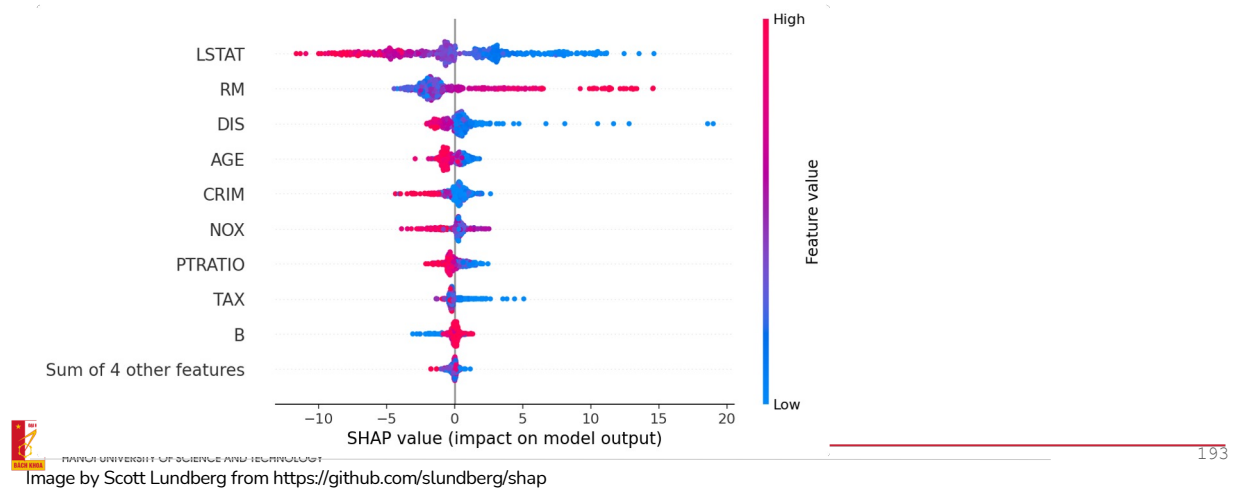
Image by Scott Lundberg from <https://github.com/slundberg/shap>

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SHAP: SHapley Additive exPlanations

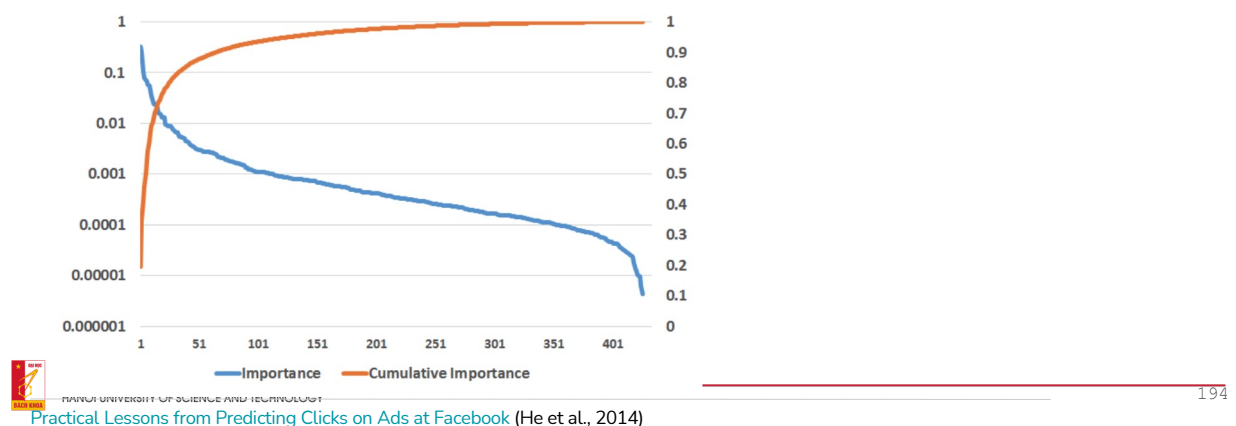
- Measuring a feature's contribution to **the entire model**



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Measuring feature importance @ Facebook

- Top 10 features: 50% total feature importance
- Bottom 300 features: <1% total feature importance



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<https://github.com/interpretml/interpret>

```
ebm = ExplainableBoostingClassifier()  
I
```

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Feature engineering: the more the better?

- Adding more features tends to improve model performance

How can having too many features be bad?



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Too many features can be bad ...

- Training:
 - Overfitting
 - More features, more opportunity for data leakage



Too many features can be bad ...

- Training:
 - Overfitting
 - More features, more opportunity for data leakage
- Inference
 - Increase inference latency with online prediction
 - Might cause increased memory usage -> more expensive instance required



Too many features can be bad ...

- Training:
 - Overfitting
 - More features, more opportunity for data leakage
- Inference
 - Increase inference latency with online prediction
 - Might cause increased memory usage -> more expensive instance required
- Stale features become technical debts
 - E.g. if zip codes are no longer allowed for predictions, all features that use zip codes will need to be updated



Too many features can be bad ...

- Training:
 - Overfitting
 - More features, more opportunity for data leakage
- Inference
 - Increase inference latency with online prediction
 - Might cause increased memory usage -> more expensive instance required
- Stale features become technical debts

Solution:

- Clean up stale / ineffective features
- Store features in case you want to reuse them
 - Feature management



9 best practices for feature engineering

1. Split data by time instead of doing it randomly.
2. If you oversample your data, do it after splitting.
3. Use statistics/info from the train split, instead of the entire data, for feature engineering: scaling, normalizing, handling missing values, creating n-gram count, item encoding, etc.
4. Understand how your data is generated, collected, and processed. Involve domain experts if necessary.
5. Keep track of data lineage.
6. Understand feature importance to your model.
7. Measure correlation between features and labels.
8. Use features that generalize well.
9. Remove stale features from your models.

Model selection

ML algorithm

- Function to be learned
 - E.g. model architecture, number of hidden layers
- Objective function to optimize (minimize)
 - Loss function
- Learning procedure (optimizer)
 - Adam, Momentum



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6 tips for evaluating ML algorithms



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1. Avoid the state-of-the-art trap

- SOTA's promise
 - Why use an old solution when a newer one exists?
 - It's exciting to work on shiny things
 - Marketing



Replying to @chipro

This is how every conversation went when someone present the SOTA Transformer in a meeting with stakeholders.



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1. Avoid the state-of-the-art trap

- SOTA's reality
 - SOTA on research data != SOTA on your data
 - Cost
 - Latency
 - Proven industry success
 - Community support



Replying to @chipro

This is how every conversation went when someone present the SOTA Transformer in a meeting with stakeholders.



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2. Start with the simplest models

- Easier to deploy
 - Deploying early allows validating pipeline
- Easier to debug
- Easier to improve upon



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2. Start with the simplest models

- Easier to deploy
 - Deploying early allows validating pipeline
- Easier to debug
- Easier to improve upon
- Simplest models != models with the least effort
 - BERT is easy to start with pretrained model, but not the simplest



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3. Avoid human biases in selecting models

- A tale of human biases
 - Papers proposing LSTM variants show that the variants improve upon the vanilla LSTM.
 - Do they?



3. Avoid human biases in selecting models

- A tale of human biases
 - Papers proposing LSTM variants show that the variants improve upon the vanilla LSTM.
 - Do they?

LSTM: A Search Space Odyssey

Klaus Greff, Rupesh K. Srivastava, Jan Koutník, Bas R. Steunebrink, Jürgen Schmidhuber

We conclude that the most commonly used LSTM architecture (vanilla LSTM) performs reasonably well on various datasets. None of the eight investigated modifications significantly improves performance.

3. Avoid human biases in selecting models

- It's important to evaluate models under comparable conditions
 - It's tempting to run more experiments for X because you're more excited about X
- Near-impossible to make blanket claims that X is *always* better than Y



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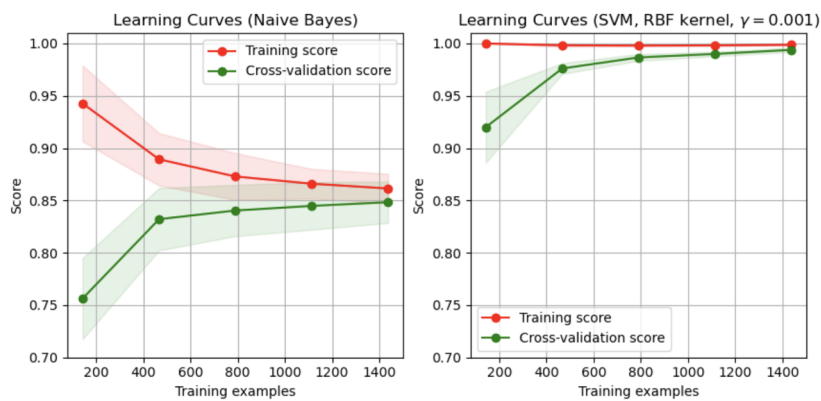
Better now vs. better later

- Best model **now** != best model **in 2 months**
 - Improvement potential with more data
 - Ease of update



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Learning curve



Good for estimating if performance can improve with more data



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https://scikit-learn.org/stable/auto_examples/model_selection/plot_learning_curve.html

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5. Evaluate trade-offs

- False positives vs. false negatives
- Accuracy vs. compute/latency
- Accuracy vs. interpretability



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6. Understand your model's assumption

- IID
 - **Neural networks** assume that examples are **independent and identically distributed**
- Smoothness
 - **Supervised algorithms** assume that there's a set of functions that can transform inputs into outputs such that **similar inputs are transformed into similar outputs**
- Tractability
 - Let X be the input and Z be the latent representation of X . **Generative models** assume that it's tractable to compute $P(Z|X)$.
- Boundaries
 - **Linear classifiers** assume that **decision boundaries are linear**.
- Conditional independence
 - **Naive Bayes classifiers** assume that the **attribute values are independent of each other given the class**.



6 tips for evaluating ML algorithms

1. Avoid the state-of-the-art trap
2. Start with the simplest models
3. Avoid human biases in selecting models
4. Evaluate good performance now vs. good performance later
5. Evaluate trade-offs
6. Understand your model's assumptions



Ensembles



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Ensemble

- Creating a strong model from an ensemble of weak models

Base learners



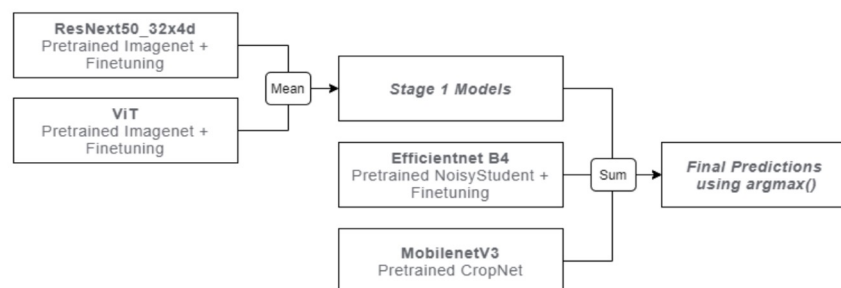
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Ensembles: extremely common in leaderboard style projects

- 20/22 winning solutions on Kaggle in Jan - Aug 2021 use ensembles
- One solution uses 33 models!



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Why does ensembling work?

- Task: email classification (SPAM / NOT SPAM)
- 3 uncorrelated models, each with accuracy of 70%
- Ensemble: majority vote of these 3 models
 - Ensemble is correct is at least 2 models are correct

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Why does ensembling work?

- 3 models, each with 70% accuracy
- Ensemble is correct if at least 2 models are correct
- Probability at least 2 models are correct: $34.3\% + 44.1\% = 78.4\%$

Outputs of 3 models	Probability	Ensemble's output
All 3 are correct	$0.7 * 0.7 * 0.7 = 0.343$	Correct
Only 2 are correct	$(0.7 * 0.7 * 0.3) * 3 = 0.441$	Correct
Only 1 is correct	$(0.3 * 0.3 * 0.7) * 3 = 0.189$	Wrong



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Why does ensembling work?

- 3 models, each with 70% accuracy
- Ensemble is correct if at least 2 models are correct
- Probability at least 2 models are correct: $34.3\% + 44.1\% = 78.4\%$

- The less correlation among base learners, the better
- Common for base learners to have different architectures



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Ensemble

- Bagging
- Boosting
- Stacking



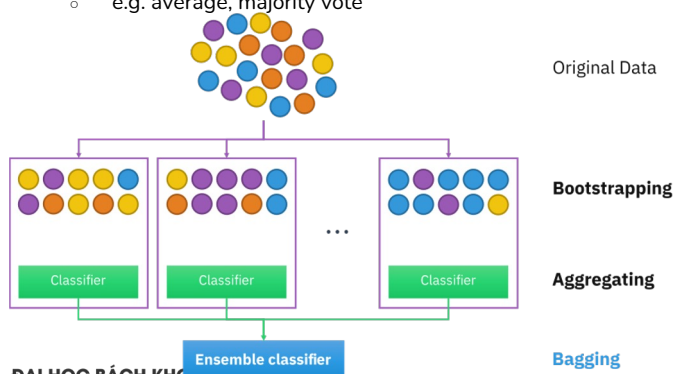
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Bagging

- Sample **with replacement** to create different datasets
- Train a classifier with each dataset
- Aggregate predictions from classifiers
 - e.g. average, majority vote



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Illustration by [Sirakorn](#)

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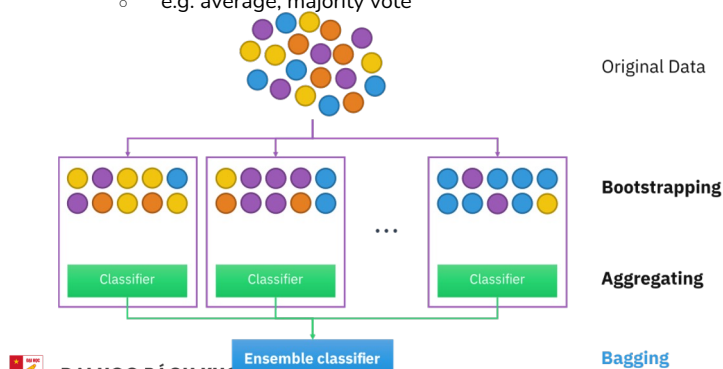
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Bagging

- Sample **with replacement** to create different datasets
- Train a classifier with each dataset
- Aggregate predictions from classifiers
 - e.g. average, majority vote

- Generally improves unstable methods e.g. neural networks, trees
- Can degrade stable methods e.g. kNN

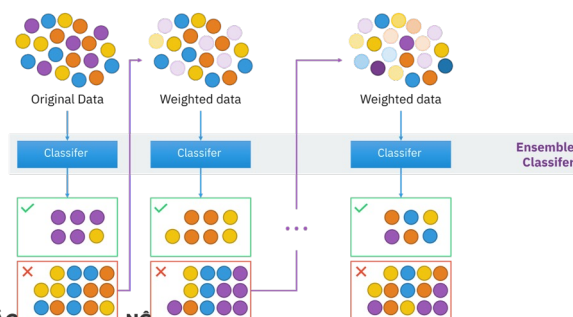
[Bagging Predictors](#) (Leo Breiman, 1996)



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Boosting

1. Train a weak classifier
2. Give samples misclassified by weak classifier higher weight
3. Repeat (1) on this reweighted data as many iterations as needed
4. Final strong classifier: weighted combination of existing classifiers
 - a. classifiers with smaller training errors have higher weights



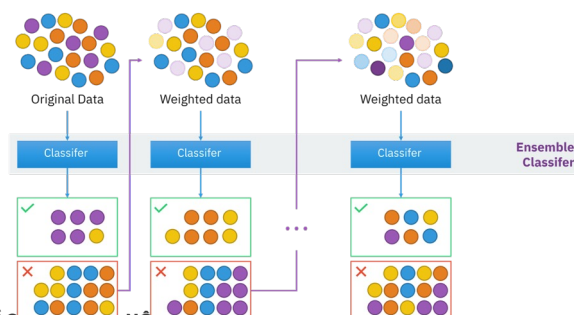
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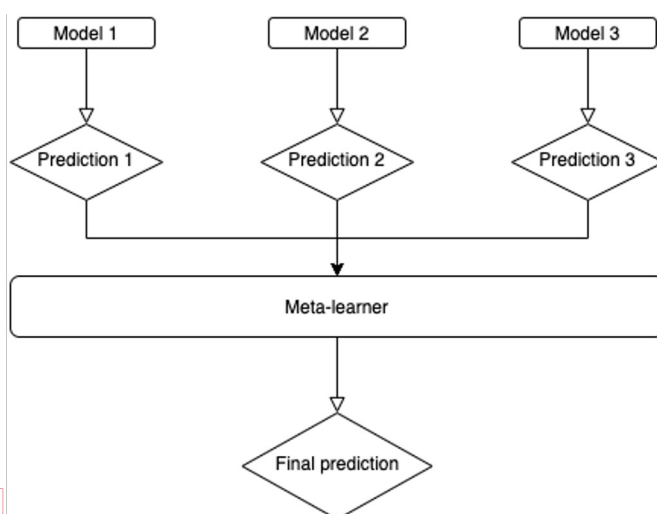
Extremely popular:

- XGBoost
- LightGBM



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Stacking



Majority vote, logistic regression, simple NN

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AutoML



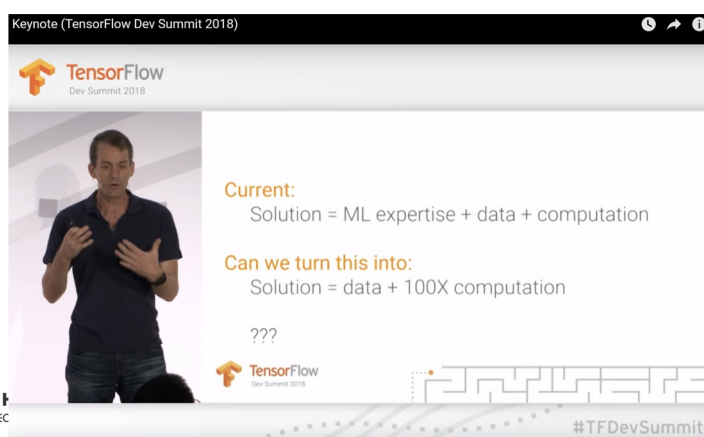
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AutoML

- A good ML researcher is someone who will automate themselves out of job
- Google: what if we replace ML experts with 100x compute?



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AutoML

- Soft AutoML:
 - hyperparameter tuning
- Hard AutoML
 - neural architecture search
 - learned optimizer

More computationally
expensive




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Soft AutoML: Hyperparameter tuning

- Weaker models with well-tuned hyperparameters can outperform fancier models
 - [On the State of the Art of Evaluation in Neural Language Models](#) (Melis et al. 2018)



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Soft AutoML: Hyperparameter tuning

- Many hyperparameters to tune

```
model_type = "bert"

def __init__(
    self,
    vocab_size=30522,
    hidden_size=768,
    num_hidden_layers=12,
    num_attention_heads=12,
    intermediate_size=3072,
    hidden_act="gelu",
    hidden_dropout_prob=0.1,
    attention_probs_dropout_prob=0.1,
    max_position_embeddings=512,
    type_vocab_size=2,
    initializer_range=0.02,
    layer_norm_eps=1e-12,
    pad_token_id=0,
    position_embedding_type="absolute",
    use_cache=True,
    classifier_dropout=None,
    **kwargs
):
```



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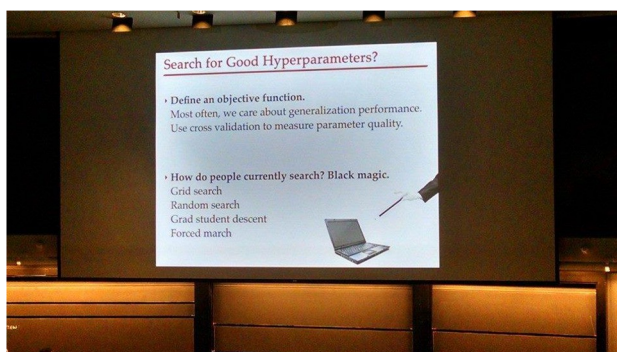
https://github.com/huggingface/transformers/blob/master/src/transformers/models/bert/configuration_bert.py

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Soft AutoML: Hyperparameter tuning

- Graduate Student Descent (GSD)
 - A graduate student fiddles around with the hyperparameters until the model works



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<https://twitter.com/GuyZys/status/592847074170896384/photo/1>

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Soft AutoML: Hyperparameter tuning

- Hyperparam tuning has become a standard part of ML workflows
- Built-in with frameworks
 - TensorFlow: Keras Turner
 - scikit-learn: auto-sklearn
 - Ray Tune
- Popular algos:
 - Random search
 - Grid search
 - Bayesian optimization



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NAS: Neural architecture search

- Search space
 - Set of operations
 - e.g. convolution, fully-connected, pooling
 - How operations can be connected

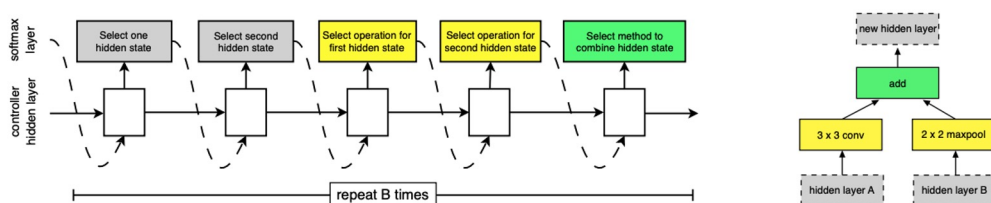


Figure 3. Controller model architecture for recursively constructing one block of a convolutional cell. Each block requires selecting 5 discrete parameters, each of which corresponds to the output of a softmax layer. Example constructed block shown on right. A convolutional cell contains B blocks, hence the controller contains $5B$ softmax layers for predicting the architecture of a convolutional cell. In our experiments, the number of blocks B is 5.



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NAS: Neural architecture search

- Search space
- Performance estimation strategy
 - How to evaluate **many** candidate architectures?
 - Ideal: should be done without having to re-construct or re-train them from scratch.



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NAS: Neural architecture search

- Search space
- Performance estimation strategy
- Search strategy
 - Random
 - Reinforcement learning
 - reward the choices that improve performance estimation
 - Evolution
 - mutate an architecture
 - choose the best-performing offsprings
 - so on

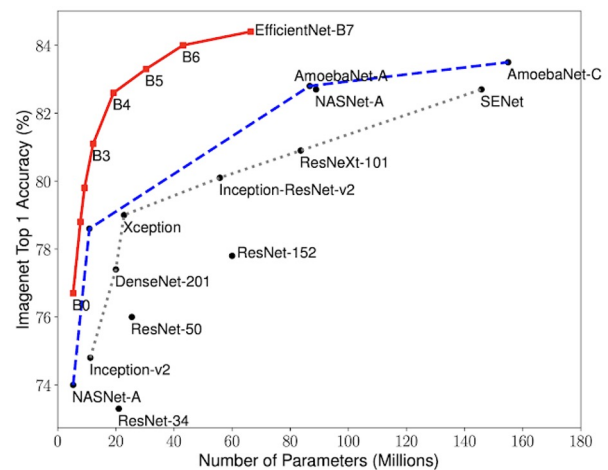


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NAS: Neural architecture search

- Search space
- Performance estimation strategy
- Search strategy

Very successful



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[EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks](#) (Tan et Le., 2019)

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Learning: architecture + learning algorithm

- Learning algorithm:
 - A set of functions that specifies how to update the weights.
 - Also called **optimizers**
 - Adam, Momentum, SGD



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Learned optimizer

Deep learning

engineering features

SIFT (Lowe et. al. 1999)
HOG (Dalal et. al. 2005)

learning features

LeNet (LeCun et. al. 1998)
AlexNet (Krizhevsky et. al. 2012)

Meta learning

engineering to learn

SGD (Robbins et. al. 1951, Bottou 2010)
Autoencoders (Hinton et. al. 2006)

learning to learn

Learning To Learn (Hochreiter et. al. 2001)
Learned Optimizers (Andrychowicz et. al.
2016, Li et. al. 2016, Wichrowska et. al.
2017, Metz et. al. 2018, 2019)



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Learned optimizer

- Learn how to learn on a set of tasks
- Generalize to **new tasks**



Using a thousand optimization tasks to learn hyperparameter search strategies



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Learned optimizer

- Learn how to learn on a set of tasks
- Generalize to **new tasks**
- The learned optimizer can then be used to train a better version of itself!



Using a thousand optimization tasks to learn hyperparameter search strategies